The Intergenerational Welfare Cost of Social Security Insolvency

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Job Market Paper

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August 2026

This paper asks which generations bear the welfare cost of Social Security insolvency. I fix OASI reserve depletion at the end of 2033, but each generation meets the same event at a different age, with different years of payments at stake and different means of adjusting. In a life-cycle model with endogenous saving, labor supply, human capital, and claiming, I quantify how the expected duration of reduced payments shifts that welfare incidence. The main result is that the cohort losing the most unpaid dollars is not the cohort losing the most welfare. Households that expect the shortfall self-insure through saving, employment, and human-capital investment, but the welfare consequence of doing so depends on age. The 2005 cohort can adjust over a longer horizon, whereas the 1975 cohort makes its adjustments closer to retirement and, under permanent reduced payments, is worse off than if the shortfall had been a surprise. Duration matters as well. When the shortfall is brief, the burden lands mainly on cohorts at or near claiming; when it is prolonged, the burden shifts toward the working generations behind them.

Keywords: OASI reserve depletion, retirement, intergenerational incidence

JEL Classification: H55, J26, E21, D15.

I am deeply grateful to my advisor, Juan Carlos Córdoba, for invaluable advice, guidance, and encouragement. I am grateful to my dissertation committee, Rajesh Singh, Quinn Weninger, Sergio Lence, and Jian Li, for their continued support. I also thank Kathleen McGarry, John Winters, Agnitra Roy Choudhury, and Xiaobo Yu for thoughtful comments and feedback, as well as participants at the 2026 North American Summer Meeting of the Econometric Society, the 2026 WEAI Annual Conference, and the Iowa State Micro/Macro Reading Group.

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1. Introduction

This paper asks how, conditional on a common OASI reserve-depletion date, the expected duration of reduced payments changes welfare incidence across birth cohorts. OASI reserve depletion, fixed in the model experiments at the end of 2033, is a single event in calendar time, but every cohort meets it at a different age. The current debate over insolvency is based largely on an accounting exercise—how many dollars of scheduled payments go unpaid, and to whom—yet accounting alone cannot identify which cohort bears the largest welfare cost because it holds behavior fixed across the event that changes behavior. No market insures against a policy-induced payment shortfall, so households must self-insure through their own choices. Age governs both sides of that problem: the options still available to a household and the number of reduced-payment years that may lie ahead.

The main idea of this paper is to simulate different birth cohorts confronting the same anticipated reserve-depletion event inside a life-cycle model in which saving, labor supply, human-capital investment, and the claiming date are endogenous choices. Two forces work against each other. The earlier reserve depletion occurs in a cohort's life, the more of its retirement is exposed to reduced payments. But an earlier event also leaves more time to adjust. Which force dominates, and by how much, is ultimately a quantitative question that accounting alone cannot answer. The central finding is that the generation with the largest accounting loss need not be the generation whose lifetime welfare falls the most.

Empirics. I begin with two pieces of empirical evidence about perceived Social Security benefit risk. First, beliefs about future payments are widely dispersed across households, consistent with Luttmer and Samwick (2018), and are associated with intended work at older ages. In the 2006 Health and Retirement Study (HRS), respondents ages 50–61 assign an average probability of 60.8 percent that a policy change within ten years will reduce their own future payments. A one-standard-deviation higher perceived risk is associated with a 2.3-percentage-point higher reported probability of working full time after age 62. This relationship is descriptive, not causal.

The follow-up evidence draws a further distinction between beliefs, plans, and realized work. Among respondents observed again in 2010, a 10-percentage-point higher stated probability of working after age 62 is associated with a 3.7-percentage-point higher probability of full-time work, conditional on prior full-time status and household characteristics.

Stated plans therefore have predictive content for later employment. The direct coefficients on perceived personal benefit risk in the later-employment regressions, however, have confidence intervals that include zero. I use this evidence only to motivate a model in which benefit beliefs are associated with choices made before payments change; the estimated relationship is neither a calibration target nor evidence that perceived risk causes realized employment.

Second, public information about OASI finances has changed over calendar time. I construct a monthly OASI financial-risk index from four blocks: the projected depletion date and actuarial deficit reported in Trustees Reports, labor-market conditions related to payroll revenue, revisions across successive Trustees Reports, and news-based economic-policy uncertainty (Baker, Bloom, and Davis 2016). Because an index score is not a probability, I map the annual index state to HRS survey means for the perceived chance that Social Security becomes less generous. The two mapping parameters are fit by weighted nonlinear least squares, using each survey wave's sample share as its weight. The fitted annual onset hazard lies between 9.1 and 10.2 percent through 2007, averages 12.2 percent from 2010 through 2019, and reaches 14.1 percent in the partial 2026 data. This series is a survey-based proxy for perceived broad benefit risk, not a direct measure of an objective OASI depletion probability.

Together, the empirical results show that benefit beliefs are dispersed, are associated with stated work plans, and evolve with public information. They do not show how much of a payment shortfall households offset through saving, labor supply, human capital, and claiming, or how the same calendar event translates into welfare at different ages. Answering those questions requires a structural life-cycle model in which these adjustment margins are endogenous.

Model. I extend the life-cycle model of Fan, Seshadri, and Taber (2024). Households choose consumption, saving, binary full-time employment, Ben-Porath human-capital investment, and when to claim Social Security. Employment and training affect current earnings, future wages, and covered earnings, so households can respond both before and after OASI reserve depletion rather than holding earnings fixed. The aggregate payment regime is an exogenous state, not a household choice. Its current realization changes current OASI payments, benefit taxation, disposable resources, and the budget constraint, while its transition hazards enter continuation values.

The experiment compares three matched environments. In the full-payment benchmark, households perceive no onset risk and receive full scheduled payments. In the surprise-shortfall environment, they use the full-payment policy through the end-of-2033 event, then receive 78 percent of scheduled payments beginning in 2034 and reoptimize. In the anticipated-shortfall environment, households make choices using cohort-specific onset beliefs but experience the same objective payment histories as surprised households. These environments support three comparisons: surprise shortfall relative to full payment, anticipation relative to surprise, and anticipated shortfall relative to full payment. The underlying utility differences telescope, but each consumption-equivalent variation (CEV) is a separate nonlinear comparison.

The permanent reduced-payment boundary illustrates the three comparisons. The surprise-shortfall welfare cost rises from 0.229 percent of lifetime consumption for the 1957 cohort to 0.756 percent for the 1965 cohort, 1.157 percent for the 1975 cohort, 1.194 percent for the 1982 cohort, and 1.334 percent for the 2005 cohort. This comparison includes reoptimization after the surprise and is therefore a model-implied consequence of the shortfall, not a mechanical exposure measure.

The anticipation comparison holds the objective payment histories fixed and captures the welfare consequence of choices made under perceived onset risk. Its sign varies across cohorts. Under permanent reduced payments, anticipation raises the estimated welfare cost of the 1975 cohort by 0.104 percentage points and that of the 1982 cohort by 0.070 percentage points, although the latter conditional Monte Carlo interval includes zero. It lowers the estimated cost of the 2005 cohort by 0.075 percentage points. The estimates for the 1957 and 1965 cohorts are close to zero, with conditional intervals that include zero. The model profiles show the economic trade-off behind these comparisons: anticipatory saving, employment, human-capital investment, and claiming choices can support consumption after payments fall, but they can also require earlier sacrifices of consumption or leisure.

The total comparison evaluates anticipated shortfall directly against full payment. Under permanent reduced payments, the total welfare cost is 0.180 percent for the 1957 cohort, 0.747 percent for the 1965 cohort, and 1.257, 1.261, and 1.261 percent for the 1975, 1982, and 2005 cohorts. These results show why unpaid dollars and welfare can rank cohorts differently: the CEV values the complete model-implied allocation, including consumption, leisure, and endogenous choices, whereas accounting values missing transfers along a fixed allocation.

Model Experiment. The primary model experiment changes only the expected duration of the reduced-payment state. For every finite duration, full-payment restoration arrives with a constant annual hazard, so the realized duration remains uncertain even when its expectation is fixed. I consider expected durations of one, two, three, five, ten, twenty, and forty years and also report permanent reduced payments, the zero-restoration-hazard boundary.

The full-payment benchmark and reduced-payment paths are imposed exogenously. In particular, the model does not specify the tax or spending changes that would finance restored full scheduled payments, and households are not charged for that financing. The estimates should therefore be read as conditional, partial-equilibrium welfare incidence along an exogenous payment-duration frontier, not as a complete welfare evaluation of a legislative reform.

Expected duration changes both the size of the welfare costs and which cohort bears the largest cost. At an expected duration of five years, the 1965 cohort has the largest total cost, 0.262 percent, compared with 0.137 percent for the 1957 cohort and 0.122 percent for the 1975 cohort. At twenty years, the point estimate for the 1975 cohort exceeds that for the 1965 cohort, 0.668 versus 0.619 percent. Under permanent reduced payments, the three youngest cohorts each lose approximately 1.26 percent of lifetime consumption, compared with 0.75 percent for the 1965 cohort and 0.18 percent for the 1957 cohort. Thus, a brief shortfall concentrates the modeled burden among cohorts at or near the claiming transition, whereas greater persistence shifts the largest modeled cost toward younger cohorts.

Two mechanisms organize this rotation. Fixed-behavior benefit exposure records how much of a cohort's scheduled OASI stream lies in reduced-payment years. Adjustment capacity describes the saving, employment, human-capital, and claiming margins available when that exposure arrives. The surprise-shortfall CEV rises with duration for every cohort, especially for younger cohorts as more of their retirement years enter the reduced-payment state. The anticipation CEV can have either sign because choices that support consumption after depletion can be costly before depletion. The total cost can therefore be non-monotonic even as objective exposure rises.

The comparison with fixed-behavior accounting reinforces the distinction between missing transfers and welfare. The accounting ranking changes with the date to which unpaid payments are discounted. Even after both measures are placed on a lifetime-consumption base, the permanent-case CEV is 15–38 percent below the accounting loss. This gap compares two valuation concepts; it is not a behavioral decomposition or a measure of income "recovered" through adjustment.

All duration results use one canonical simulation with common random numbers across the three matched environments and 20,000 simulated people per cohort–duration cell. The reported intervals are paired conditional Monte Carlo intervals. They measure finite-simulation variation with calibrated parameters, empirical inputs, and the model specification held fixed. They do not incorporate estimation uncertainty, sampling uncertainty in the empirical inputs, model uncertainty, or multiple-comparison adjustments and should not be read as general statistical-significance tests.

Related Literature. This paper makes two contributions. First, it turns OASI reserve depletion from an accounting question into an intergenerational welfare-incidence question and contributes to the literature on Social Security policy risk and uncertainty. Shoven and Slavov (2006) compare political risk to scheduled benefits with the market risk of a funded system. Luttmer and Samwick (2018) elicit subjective distributions of future Social Security payments and quantify the welfare cost of perceived uncertainty. Caliendo, Gorry, and Slavov (2019) study uncertainty about when a reform will arrive, while Kitao (2014, 2018) quantify the intergenerational effects of alternative reform paths. I ask a complementary question: conditional on one depletion date and an exogenous payment path, how does expected reduced-payment duration change welfare incidence across cohorts?

Second, the paper models reserve depletion as a common calendar-time event that each cohort encounters at a different age and separates the shortfall comparison from the anticipation comparison. Structural models of retirement, labor supply, and claiming (Rust and Phelan 1997; French 2005; French and Jones 2011; Rogerson and Wallenius 2013; Goda et al. 2018; Shoven, Slavov, and Wise 2017; Bairoliya and McKiernan 2026; Pashchenko and Porapakkarm 2024; Fan, Seshadri, and Taber 2024) typically take Social Security rules as known, while reform studies such as Kitao (2014); Jones and Li (2023) impose policy paths chosen by the researcher. Here, a common subjective calendar distribution becomes a cohort-specific onset hazard in age time. The endogenous adjustment margins connect the analysis to work on precautionary saving (Gourinchas and Parker 2002; Cagetti 2003), Social Security wealth and public annuities (Geanakoplos and Zeldes 2010; Caliendo, Guo, and Hosseini 2014; Hosseini 2015), and expectations under policy uncertainty (Bloom 2009;

Baker, Bloom, and Davis 2016; Jurado, Ludvigson, and Ng 2015; Caldara and Iacoviello 2022).

Roadmap. Section 2.1 documents perceived personal benefit risk and intended work. Section 2.2 constructs the OASI financial-risk index and maps public information into a survey-based onset hazard. Section 3 presents the life-cycle model and its payment regimes. Section 4 defines the counterfactual environments and welfare comparisons. Section 5 traces welfare incidence along the expected-duration frontier, and Section 6 concludes.

2. Perceived Social Security Benefit Risk

This section presents two pieces of motivating evidence linking perceived Social Security benefit risk to stated economic choices. Direct evidence on OASI reserve depletion is unavailable because depletion has not occurred, and no market insurance protects households against a policy-induced decline in future payments.[1] I therefore use survey questions about the possibility that Social Security will become less generous. These questions measure broad policy risk rather than the structural model's specific reserve-depletion event (Diamond 1997; Shoven and Slavov 2006; Luttmer and Samwick 2018; Caliendo, Gorry, and Slavov 2019).

2.1. Perceived Benefit Risk and Intended Work

Do people approaching retirement report a greater intention to work when they perceive more risk to their own future Social Security payments? This relationship provides descriptive evidence for the model's premise that benefit beliefs are associated with choices made before payments change.

I use the 2006 Health and Retirement Study (HRS) Internet Survey. I restrict the sample to respondents ages 50–61 who answered the relevant questions, are not receiving Social Security, and expect to receive benefits in the future. These respondents have not yet reached the earliest retirement-benefit claiming age of 62. The age-62 regression sample contains 3,359 individuals. Appendix A reports summary statistics. The reported sample

[1]A life annuity insures idiosyncratic longevity risk; it does not insure the level of Social Security payments.

means and regressions are unweighted and therefore describe this Internet Survey sample rather than a nationally representative population.

The main explanatory variable is *perceived personal benefit risk*, denoted by R_i^P.[2] It is the respondent's reported probability that Social Security will change during the next ten years in a way that lowers the respondent's own payments relative to current law. Its sample mean is 60.8 percent, with a standard deviation of 28.2 percentage points. I do not call this measure insolvency risk or a subjective depletion hazard because the survey permits many policy mechanisms.

The main outcome is the individual's reported probability of working full time after age 62, denoted by $W_{i,62}$. It measures intended work at an older age, not realized employment. Its sample mean is 47.2 percent and its standard deviation is 37.8 percentage points. The dispersion partly reflects round-number heaping at 0, 50, and 100, a common feature of subjective-probability questions in household surveys (Manski 2004; Hurd 2009).

I also construct *perceived general program risk*, R_i^G, from the reported probability that Social Security will become less generous in general. This separates the risk a person assigns to personal payments from a belief about program-wide generosity. Some respondents may expect a future reform but also expect Congress to protect people close to retirement. Holding R_i^G fixed, the coefficient below compares people with similar views about the program but different perceived risks to their own payments.

I estimate the regression equation

$$W_{i,62} = \alpha + \beta R_i^P + \theta R_i^G + X_i'\gamma + \varepsilon_i. \tag{1}$$

The controls X_i include age fixed effects, gender, education, race and ethnicity, marital status, self-reported health, earnings, and nonhousing wealth. Standard errors are clustered by household, allowing responses within a household to be correlated. Both risk measures are expressed in units of 10 percentage points. The coefficient β therefore measures the conditional difference in the reported work probability associated with a 10-percentage-point difference in perceived personal benefit risk.

[2]The general-risk question (HRS 2006, question P110) reads: "Thinking of the Social Security program in general and not just your own Social Security benefits: On a scale from 0 to 100, what is the percent chance that Congress will change Social Security sometime in the next 10 years, so that it becomes less generous than now?" The personal-risk question follows it: "We just asked you about changes to Social Security in general. Now we would like to know whether you think these Social Security changes might affect your own benefits." The question does not identify a specific mechanism, such as a statutory entitlement change, a higher retirement age, benefit taxation, or reserve depletion.

Figure 1 shows the result. A 10 percentage-point difference in perceived personal benefit risk is associated with a 0.83 percentage-point difference in the reported probability of working after age 62. A difference of one standard deviation in perceived risk—28.2 percentage points—therefore corresponds to about a 2.3 percentage-point difference in the work probability. This difference is about 5 percent of the sample's average work probability. By comparison, the coefficient on general program risk is -0.32 (standard error 0.32), and its confidence interval includes zero.

The unadjusted survey means point in the same direction. Among people who report a 0–24 percent chance that policy changes will lower their personal payments, the average probability of working after age 62 is 44.9 percent. Among those who report a 75–100 percent chance, the average is 48.9 percent. The average rises across all four risk groups. Because many respondents give round-number answers such as 0, 50, or 100, I treat this panel only as a descriptive group comparison. The regression panels report the conditional relationship after including the controls.

The estimated relationship remains similar in the alternative specifications reported in Table A.2. It changes little when I add the respondent's expected chance of survival and perceived chance of a major economic depression, use controls measured before the belief question, or include pension coverage on the current job. Pension coverage is undefined for respondents who are not working, so I code it as no pension, has a pension, or not currently working. With this control, the coefficient on perceived personal benefit risk is 0.84 (standard error 0.27), close to the baseline 0.83. When the outcome is the reported probability of working after age 65, the coefficient is 0.77 (standard error 0.25), or 0.74 (standard error 0.25) with the pension control.

I next examine whether stated work plans contain information about subsequent employment. I follow 1,254 respondents who were ages 58–61 in 2006 to the 2010 HRS, by which time all had reached age 62. The final linear-probability-model sample contains 1,119 respondents with observed full-time employment and nonmissing controls.

A 10-percentage-point higher stated probability of working after 62 in 2006 is associated with a 3.7-percentage-point higher probability of full-time work in 2010 (standard error 0.4), conditional on 2006 full-time status and the same household characteristics. Stated plans therefore have predictive content for later employment. By contrast, the coefficients on perceived personal benefit risk in the 2008 and 2010 full-time employment regressions have confidence intervals that include zero. Table A.3 reports these regressions and the

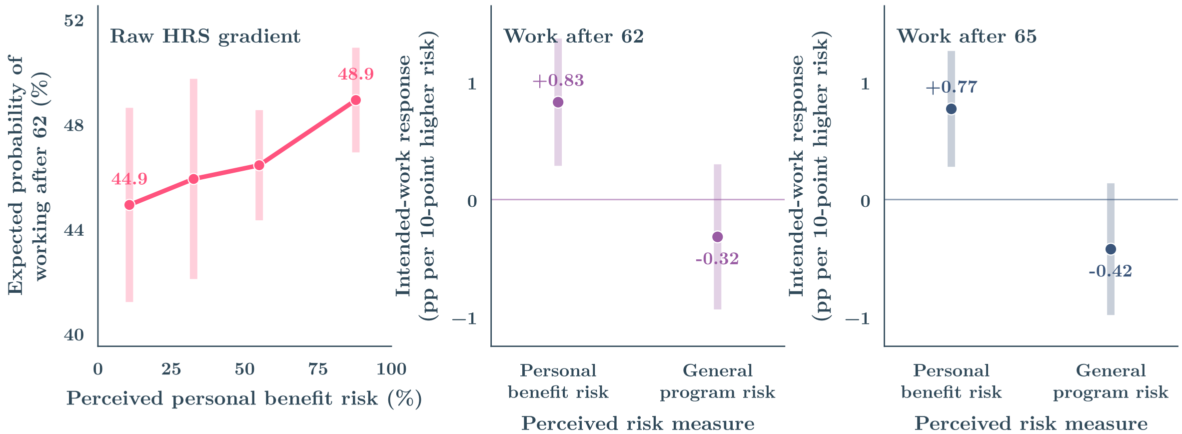

FIGURE 1. Perceived Social Security risk and intended work at older ages

Note. The sample consists of people ages 50–61 who answered the 2006 HRS survey for themselves, are not receiving Social Security, expect to receive benefits in the future, and answered the questions about personal and general risk. The left panel shows the unadjusted average for each perceived-personal-benefit-risk group. The middle and right panels show coefficients from equation (1), estimated separately for the reported probability of working after ages 62 and 65. The bars show 95 percent confidence intervals. The age-62 regression includes 3,359 people, and the age-65 regression includes 3,362. Standard errors are clustered by household.

analogous 2008 exercise.

The evidence supports a limited conclusion. Respondents who perceive their own future benefits as less secure report a greater probability of working at older ages, and stated work plans predict later full-time employment. The regressions do not establish that benefit beliefs cause either planned or realized work. Stable employment preferences, financial knowledge, political beliefs, and private information about health or wealth may affect both beliefs and work plans. The survey question also combines OASI reserve-depletion risk with other policy changes that could reduce benefits. I therefore treat β as a descriptive association. It is not a model calibration target and does not enter the welfare calculations.

2.2. Mapping Public Information into Subjective Onset Hazards

The cross-sectional evidence in Section 2.1 does not show how public information about Social Security finances has changed over time. I therefore construct an *OASI financial-risk index*. The index is a summary of public information, not a probability. I then use HRS survey answers to map the index into an empirical series of subjective onset hazards. This series measures how the perceived risk of entering a reduced-payment state could evolve with observable financial conditions; the main structural counterfactuals use the

Trustees-informed calendar distribution specified in Section 4.

Households observe public information about Social Security's finances and the economy that supports those finances. I combine that information into four blocks. The OASI-finance block, weighted 0.30, measures the size and proximity of the financing problem. The macro-stress block, weighted 0.20, measures labor-market conditions because covered earnings generate payroll-tax revenue. The official-uncertainty block, weighted 0.10, measures revisions between successive Trustees Reports. The news-based Economic Policy Uncertainty index of Baker, Bloom, and Davis (2016), weighted 0.40, measures broader uncertainty about government economic policy. The weights are fixed and sum to one.

I build the index in three steps. First, I orient every input so that a larger number indicates greater OASI financial risk. Second, within each block I use a one-sided Kalman filter to summarize the information released by each month. "One-sided" means that a historical value uses no later information. Third, I combine the four filtered blocks using the fixed weights above. The index tracks OASI rather than combined OASDI because the structural experiment covers OASI retired-worker payments, not Disability Insurance. Appendix B lists every variable, transformation, release date, weight, and filtering rule.

Figure 2 shows the sources of movement in the headline index. The fiscal component combines OASI finances and labor-market conditions; the uncertainty component combines revisions in official forecasts with broader economic-policy uncertainty. Each component is centered at 100 and rescaled so that their weighted sum equals the headline at every date.

An index score is not a probability: a score of 120, for example, does not mean that a payment shortfall has a 120 percent chance of occurring. To obtain a probability scale, I use the 2006, 2007, and 2011 HRS Internet Surveys, which ask for the chance that Congress will make Social Security less generous within ten years. Let G_y be the calendar-year average of the unsmoothed OASI state. I estimate the subjective onset hazard

$$\lambda_y = \Lambda(\alpha_0 + \alpha_1 G_y),$$

where Λ is the logistic function. The maintained mapping interprets λ_y as the annual probability of entering a reduced-payment state in year y, conditional on not entering it earlier.

The survey asks about a ten-year horizon, whereas the model uses an annual hazard.[3] I

[3] The survey asks about broad legislated changes in generosity; the model's reduced-payment state

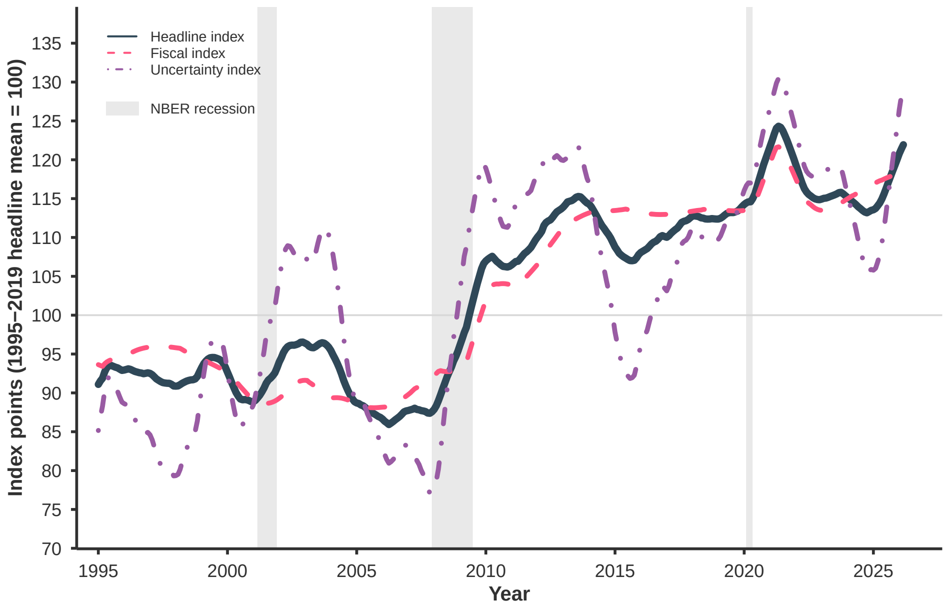

FIGURE 2. OASI financial-risk index and component indexes

Note. The fiscal index rescales the OASI-finance and macro-stress contributions, whose headline weights are 0.30 and 0.20. The uncertainty index rescales the official-uncertainty and economic-policy-uncertainty contributions, whose headline weights are 0.10 and 0.40. The displayed component indexes are normalized so that $I_t^H = 0.70 I_t^F + 0.30 I_t^U$; the 0.70 and 0.30 coefficients are display normalizations, not replacements for the four block weights. Before smoothing, all three indexes average 100 from January 1995 through December 2019, and a higher value indicates a more adverse OASI financial environment. The 2026 endpoint contains data from January through March. Shaded areas mark NBER recessions.

assume the annual hazard is constant within the survey horizon, so its ten-year counterpart is $1 - (1 - \lambda_y)^{10}$. I fit α_0 and α_1 to the three survey-wave means by weighted nonlinear least squares, using each wave's sample share as its weight. With three moments and two parameters, the fitted values approximate rather than exactly match all three means. Appendix B reports the objective, fit, and sensitivity of this mapping.

Figure 3 shows the fitted series. It is low and stable from 1995 through 2007, between 9.1 and 10.2 percent, and then rises during the financial crisis. It averages 12.2 percent from 2010 through 2019, reaches 13.7 percent in 2020, and equals 14.1 percent in the partial 2026 data. On the ten-year survey scale, the same path runs from about 64 percent in the late 1990s to 77 percent in 2020.

The two recessions move the index for different reasons. In 2009, labor-market stress and economic-policy uncertainty account for the sharp increase. Those components later

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follows OASI reserve depletion. The mapping treats the survey answers as informative about the level of perceived benefit risk without equating the two mechanisms.

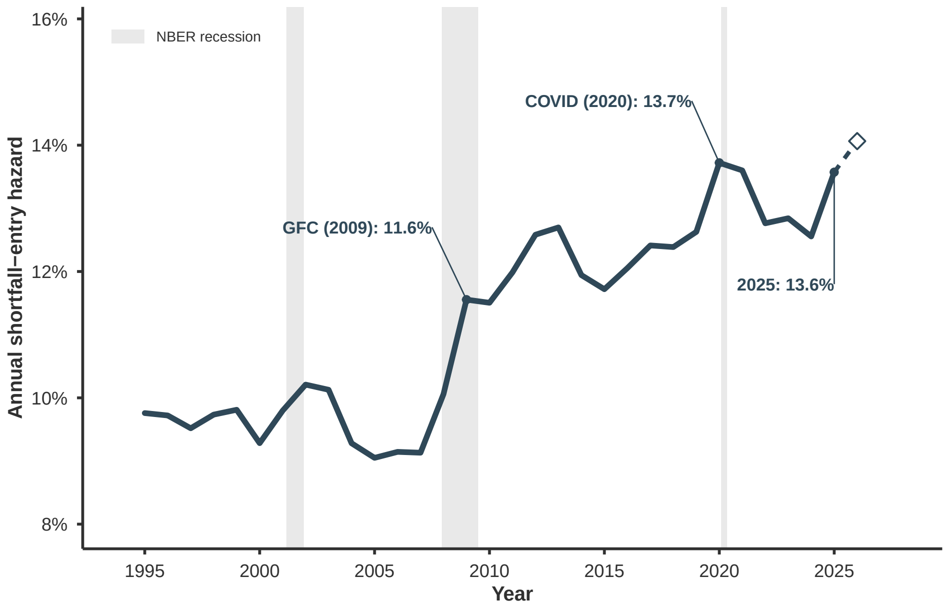

FIGURE 3. Empirical subjective onset hazard

Note. The figure maps the annual OASI financial-risk state into $\lambda_y = \Lambda(\alpha_0 + \alpha_1 G_y)$. The parameters use HRS Internet Survey averages for the 10-year probability that Social Security will become less generous. The expression $1 - (1 - \lambda_y)^{10}$ puts the annual hazard and survey answers on a common horizon. The 2026 endpoint uses data from January through March and is marked as a partial year. Shaded areas mark NBER recessions.

mean-revert, but the level remains elevated because the OASI-finance component rises as successive Trustees Reports move projected reserve depletion closer and widen the actuarial deficit. COVID-19 produces a second uncertainty spike in 2020, while the underlying upward drift remains fiscal.

Two cautions govern interpretation. First, λ_y is an annual onset hazard, not a cumulative belief. Second, the horizontal axis is calendar time, whereas households make decisions in age time. A household born in 1957 was 38 when the series begins and encountered its highest values only after age 60. A household born in 2005 turned 18 in 2023 and has encountered the elevated values throughout its working life. The same calendar-time path therefore maps into a different sequence of age-specific hazards for every birth cohort.

3. Life-Cycle Model

In this section, I extend the life-cycle model of Fan, Seshadri, and Taber (2024), in which an individual chooses whether to work, how much to consume and save, how much to invest in human capital, and when to claim Social Security. The model is partial equilibrium. The rental price of human capital, interest rate, tax schedules, claiming rules, and paid-benefit fraction are fixed across experiments. Wages remain endogenous through human-capital investment, allowing households to adjust productivity in response to lower expected Social Security payments.

The extension adds three aggregate payment states: a pre-depletion full-payment state, a reduced-payment state, and a restored full-payment state. OASI reserve depletion moves the economy from the first state to the second. Full-payment restoration moves it from the second state to the third. During the reduced-payment state, the same paid-benefit fraction applies to old and new claimants, so early claiming cannot preserve the full-payment rate.

The main object is life-cycle behavior across birth cohorts under perceived payment-shortfall risk. Different cohorts map the same calendar-time beliefs into different age-specific onset hazards. Holding preferences and all other model primitives fixed, these belief paths determine choices before reserve depletion and the resources available when reduced payments begin.

3.1. Demographics and permanent types

Time is annual. Households enter the model at age $\underline{a} = 18$ and exit at the terminal age $T = 80$. Family status $M_t \in \{0, 1, 2\}$ distinguishes three states: single or divorced, married with a nonworking spouse, and married with a working spouse. It evolves exogenously according to the age-dependent transition matrix

$$\Pr(M_{t+1} = m' \mid M_t = m) = P_t^M(m, m'), \tag{2}$$

estimated from SIPP transition frequencies by Fan, Seshadri, and Taber (2024). Family status matters through three channels. It scales the marginal utility of consumption (capturing household size), it shifts the taste for leisure, and it determines whether spousal income arrives. Conditional on $M_t = 2$, spousal earnings are drawn from a log-normal distribution with age-specific mean and variance. In the other two states they are zero. Both family status and spousal labor supply are exogenous to the household's choices.

Households also differ permanently. A permanent type is

$$\vartheta_i = (a_{0i}, \pi_i, H_{i0}), \tag{3}$$

where a_{0i} shifts the taste for leisure, π_i is efficiency in producing new human capital, and H_{i0} is initial human capital. Following Fan, Seshadri, and Taber (2024), the joint distribution of (a_{0i}, π_i) has nine mass points, and H_{i0} depends on both components plus a residual initial-condition shock. These permanent types differ in wealth, earnings, and attachment to work at the ages when payment-shortfall risk becomes salient, so their consumption-equivalent welfare effects can differ under a common onset hazard.

Health is held fixed in the benchmark: every household remains in one health state, and the model contains neither health transitions nor active SSDI or SSI receipt. Health and disability risk are therefore outside the payment-shortfall experiment.

3.2. Preferences and labor supply

Labor supply has only the extensive margin. A household works a fixed full-time schedule or does not work, and there is no hours or part-time choice. A worker may allocate part of the fixed schedule to human-capital investment, as in Fan, Seshadri, and Taber (2024), but does not choose total hours. This binary structure maps empirical employment-rate targets directly into a single policy function.

Let $p_t \in \{0, 1\}$ indicate full-time work and $\ell_t = 1 - p_t$ leisure. Flow utility is

$$u_t(c_t, p_t, d_t; M_t, a_{0i}, \varepsilon_t) = \psi_t(M_t)\frac{c_t^{1-\eta}}{1-\eta} + \exp\Big\{a_{0i} + \alpha^M(M_t) + \varepsilon_t\Big\}\,\ell_t + \nu_t d_t, \tag{4}$$

where d_t indicates initiating Social Security benefits in the current year. The first term is CRRA utility from consumption, scaled by $\psi_t(M_t)$ so that married households value consumption differently from singles. The second term is the utility of leisure. The permanent component a_{0i}, the family shifter α^M, and an independent and identically distributed innovation ε_t enter a multiplicative log-normal taste shifter, so tastes for nonwork are always positive and skewed.

The third term, $\nu_t d_t$, is a utility payoff attached to claiming retirement benefits. The payoff is age-dependent,

$$\nu_t = \nu_{62}\,\mathbf{1}\{a_t = 62\} + [\nu_{a,65} + \nu_{b,65}(a_t - 65)]\,\mathbf{1}\{65 \le a_t \le 70\}, \tag{5}$$

so that a one-time payoff ν_{62} applies at the earliest claiming age and a separate payoff applies from the normal retirement age to age 70, trending linearly at rate $\nu_{b,65}$. It captures the determinants of claiming timing outside the model, such as program awareness, framing of the statutory ages, and liquidity events. This term allows the model to reproduce the pronounced claiming peaks at ages 62 and 65 that financial incentives alone understate (Rust and Phelan 1997).

The leisure innovation $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ is observed at the start of the period, before the work decision. Since it affects utility only through the current leisure payoff, the work decision has a two-stage structure. The household first solves the value of each work branch excluding the current leisure payoff and then compares the branches given the realized shock. Let W_t^W and W_t^N denote the optimized working and nonworking values, each already maximized over consumption, claiming, and (for the working) investment. When $W_t^W > W_t^N$, working is optimal for low taste realizations, and the indifference point solves $W_t^W = W_t^N + \exp(\mu_t + \varepsilon_t)$, which yields

$$\mu_t = a_{0i} + \alpha^M(M_t), \qquad \varepsilon_t^* = \log(W_t^W - W_t^N) - \mu_t. \tag{6}$$

Work is chosen when $\varepsilon_t < \varepsilon_t^*$, so the work probability conditional on the remaining states $\widetilde{X}_t$ is

$$\Pr(p_t = 1 \mid \widetilde{X}_t) = \Phi\left(\frac{\varepsilon_t^*}{\sigma_\varepsilon}\right). \tag{7}$$

Because the shock is normal and enters exponentially, the expectation of the maximum over the two branches has a closed form,

$$\begin{aligned} \mathcal{E}_t(W_t^W, W_t^N) = \Phi(q_t) W_t^W + [1 - \Phi(q_t)] W_t^N \\ + \exp\left(\mu_t + \frac{\sigma_\varepsilon^2}{2}\right) \Phi(\sigma_\varepsilon - q_t), \qquad q_t = \frac{\varepsilon_t^*}{\sigma_\varepsilon}, \end{aligned} \tag{8}$$

where the last term is the mean of the log-normal taste shifter truncated to the nonwork region. Appendix D.1 derives this expression and its limiting case. When $W_t^W \leq W_t^N$, nonwork is optimal for every realization and the expected value is W_t^N plus the unconditional log-normal mean. The closed form integrates the shock exactly, so no quadrature is needed and the shock never enters the state vector.

Retirement is not an absorbing state in this model. "Retirement" is simply nonwork late in life, and a nonworker may return to work in a subsequent year if the value of working rises, for example because anticipated reduced payments lower the value of remaining out of work. This flexibility matters in the counterfactuals, where some households respond by returning to work.

3.3. Human capital, training, and earnings

Wages are endogenous and evolve through a Ben–Porath technology. Productive human capital is $H_t > 0$. Conditional on working, the household chooses continuously what share of its working time to devote to training rather than production. The training choice set is

$$\mathcal{I}(p_t) = \begin{cases} [0,1], & p_t = 1, \\ \{0\}, & p_t = 0. \end{cases} \tag{9}$$

A worker devotes the share I_t to human-capital production and $1 - I_t$ to current production. With $\overline{N} = 2{,}000$ annual hours converting normalized time shares into hours, $N_t^{\text{training}} = \overline{N} I_t$ and $N_t^{\text{paid}} = \overline{N}(1 - I_t)$.

The rental price of a unit of human capital is normalized to one, so the hourly wage equals the human-capital stock,

$$w_t = H_t, \tag{10}$$

and annual labor earnings are the product of hours in production and the wage,

$$y_t = \overline{N} H_t p_t (1 - I_t). \tag{11}$$

This distinction between the wage and earnings matters for measurement. By assumption, only production time is compensated, so a worker who trains receives no compensation for the fraction I_t of the year. The hourly wage reported in the data is matched to the rental value H_t, the price of one hour of the worker's skill, while annual earnings equal that wage applied to production hours alone. The two therefore differ exactly by the factor $1 - I_t$.

Human capital accumulates according to

$$H_{t+1} = (1 - \delta) H_t + p_t \, \xi_t \pi_i I_t^{\alpha_I} H_t^{\alpha_H}, \qquad \mathbb{E}[\xi_t] = 1, \tag{12}$$

where δ is depreciation, π_i is the type-specific production efficiency, and ξ_t is a mean-one log-normal innovation. New human capital is produced only while working, so nonwork implies deterministic depreciation. This technology generates a hump-shaped life-cycle wage pattern endogenously. Investment is high when the remaining working horizon is long and declines as the horizon shortens, so wages rise early in life and eventually fall as depreciation dominates. The same technology creates a feedback channel in the payment-shortfall experiments. Anticipated reduced payments can extend the optimal working horizon and thereby raise the return to human-capital investment, late-career wages, and covered earnings. The model imposes the same depreciation rate on and off the job. Training is a continuous choice, and there is no transitory or persistent wage shock apart from the human-capital innovation.

3.4. Social Security

3.4.1. Earnings record, claiming, and benefits

Social Security enters the model through an earnings record built up during working life, a one-time claiming decision, and a benefit formula with statutory adjustments. Let $s_t \in \{0, 1\}$ indicate whether benefits were initiated before year t, and let d_t indicate initiation in the current year. Claiming is a one-time, irreversible decision available between ages 62 and 70, with the feasible set

$$s_{t+1} = \max\{s_t, d_t\}, \qquad d_t \in \begin{cases} \{0\}, & s_t = 1 \text{ or } a_t < 62, \\ \{0, 1\}, & s_t = 0 \text{ and } 62 \le a_t < 70, \\ \{1\}, & s_t = 0 \text{ and } a_t = 70. \end{cases} \tag{13}$$

Claiming and work are separate decisions. A household may claim while still working, in which case the Retirement Earnings Test below withholds part of the benefit.

The benefit formula depends on both the earnings history and the claiming age, so tracking each as its own state variable would multiply the size of the state space. The model avoids this cost with a single record state R_t whose interpretation changes at claiming. Before claiming, R_t is the Average Indexed Monthly Earnings (AIME). At initiation it is converted once to the claiming-age-adjusted benefit entitlement, and after claiming the record is no longer an earnings-average state. Formally,

$$R_t = \begin{cases} \text{AIME}_t, & s_t = 0, \\ \bar{b}_t, & s_t = 1. \end{cases} \tag{14}$$

Both objects live on a common grid, so one state dimension serves both phases of life.

The statutory AIME is the monthly average of the 35 highest years of covered earnings. To keep the earnings record one-dimensional, the model uses the recursive approximation

$$\text{AIME}_{t+1} = \text{AIME}_t + \max\left\{0, \frac{\min\{y_t, 87{,}900\}}{35 \times 12} - m(a_t)\,\text{AIME}_t\right\}, \tag{15}$$

in 2004 dollars, where covered earnings are capped at the \$87,900 taxable maximum and $m(a) = \mathbf{1}\{a - \underline{a} \geq 35\}\,\text{share}_{\min}(a)$. The interpretation is that each year of covered earnings adds its thirty-fifth part to the monthly average. Once the household has more than 35 potential earnings years, a new high year displaces the lowest year in the record, whose share of AIME is the SIPP-based estimate $\text{share}_{\min}(a)$ of Fan, Seshadri, and Taber (2024). The max operator ensures that a low-earnings year never lowers the record, as in the statute. Real earnings update AIME directly, without a separate wage-index adjustment, following the approximation in Fan, Seshadri, and Taber (2024). The monthly Primary Insurance Amount (PIA) applies the progressive bend-point formula

$$\begin{aligned} \text{PIA}(R) = 0.90 \min\{R, 612\} \\ &+ 0.32 \min\{\max(R - 612, 0), 3{,}689 - 612\} \\ &+ 0.15 \max\{R - 3{,}689, 0\}, \end{aligned} \tag{16}$$

with replacement rates of 90, 32, and 15 percent below, between, and above the \$612 and \$3,689 bend points.

The model retains the historical FST claiming rules: the normal retirement age is 65 and the earliest claiming age is 62. These fixed rules are model primitives and should not be interpreted as the statutory retirement ages of the birth-cohort labels used in the counterfactuals. At initiation, the annual scheduled entitlement adjusts the PIA for the claiming age according to

$$\bar{b}_t = 12 \cdot g(a_t) \cdot \text{PIA}(R_t), \quad g(a) = \begin{cases} 1 - 0.0667(65 - a), & 62 \leq a < 65, \\ 1, & a = 65, \\ 1 + 0.06 \min\{a - 65, 4\}, & 65 < a < 70, \\ 1.24, & a \geq 70. \end{cases} \tag{17}$$

Early claiming reduces the benefit by 5/9 of one percent per month (6.67 percent per year) for up to three years, while delayed claiming earns a 6-percent-per-year credit that stops after age 69.

A claimant who keeps working before age 70 is subject to the Retirement Earnings Test, using the 1999 rules in 2004 dollars. The amount withheld is

$$\mathrm{RET}_t = \begin{cases} \min\Big\{\bar{b}_t,\ \max\{0, (y_t - 10{,}885)/2\}\Big\}, & 62 \le a_t < 65, \\ \min\Big\{\bar{b}_t,\ \max\{0, (y_t - 17{,}575)/3\}\Big\}, & 65 \le a_t < 70, \\ 0, & \text{otherwise}, \end{cases} \tag{18}$$

so one dollar of benefits is withheld per two dollars of earnings above the exempt amount before the normal retirement age, and one per three between 65 and 69. The test is not a pure tax,[4] because withheld benefits are returned actuarially through a permanently higher entitlement,

$$\bar{b}_{t+1} = \bar{b}_t \left[1 + \frac{\mathrm{RET}_t}{\bar{b}_t}\Big(0.067 \cdot \mathbf{1}\{62 \le a_t < 65\} + 0.06 \cdot \mathbf{1}\{65 \le a_t < 70\}\Big)\right], \tag{19}$$

with the ratio set to zero when $\bar{b}_t = 0$. Covered earnings no longer update AIME after claiming.

Because disability states and disability programs are inactive, the fiscal experiment covers OASI retired-worker benefits only.

The tax and benefit rules above are the 2004 federal rules combined with the 1999 earnings test, the environment facing the cohort used in estimation. They are treated as fixed model primitives. The payment-shortfall experiment changes only the fraction of each scheduled OASI payment that is actually paid.

[4] Whether the test acts as an implicit tax on work depends on how the household values the credit. For a household that values future benefits at their actuarial worth, the credit makes the test approximately neutral. Households that discount future benefits heavily, or that doubt those benefits will be paid, experience the test as an implicit tax on earnings, and many older workers respond as if it were one by reducing earnings toward the exempt amount (Friedberg 2000; Gruber and Orszag 2003). Payment-shortfall risk sharpens this logic in the present model. The credit offsets withheld payments through future entitlements that are subject to the payment factor χ, so a household that anticipates reduced payments faces a positive expected implicit tax under the test even though the statutory credit is intended to be actuarially fair.

3.4.2. Payment Regimes and Onset Beliefs

The extension introduces an aggregate payment regime $z_t \in \{F_0, L, F_1\}$. The economy begins in the pre-depletion full-payment state F_0, may enter the reduced-payment state L, and may later enter the restored full-payment state F_1. Current payments are the same in F_0 and F_1, but their transition laws differ. I hold the paid-benefit fraction in L fixed at $\phi = 0.78$ as an exogenous severity parameter. The payment factor is

$$\chi(z_t) = \begin{cases} 1, & z_t \in \{F_0, F_1\}, \\ \phi, & z_t = L, \end{cases} \qquad 0 < \phi < 1. \tag{20}$$

Let $b_t^{pre} = \max\{0, \bar{b}_t - \mathrm{RET}_t\}$ denote the scheduled payment after the earnings test. The actual OASI payment is

$$ssb_t = \chi(z_t)\, b_t^{pre}. \tag{21}$$

Three design choices deserve emphasis. First, the paid-benefit fraction applies to payments rather than entitlements. It changes neither AIME, nor the claiming adjustment, nor the stored entitlement $\bar{b}_t$. This mirrors the maintained experiment at OASI reserve depletion, in which scheduled entitlements remain defined by statute but actual payments fall below schedule. Second, the paid-benefit fraction applies equally to households that claimed before and after depletion, so claiming early cannot lock in the full-payment rate. Any claiming response in the model therefore operates through liquidity and consumption-smoothing motives and the expected return to delayed claiming. Third, the factor is applied after the earnings test and its future-entitlement credit but before the taxable-benefit calculation, so income taxes are levied on the payment actually received. The experiment covers OASI retired-worker payments. Payroll-tax rules are unchanged and the inactive SSDI/SSI block is outside the experiment.

Beliefs about payment-shortfall onset are common in calendar time but map into different ages across birth cohorts. Each cohort c therefore uses a distinct age profile of perceived hazards generated by the same calendar distribution. The subjective transition matrix $\mathcal{Q}_c$ has three parts. From the pre-depletion state F_0, cohort c perceives an end-of-year move to the reduced-payment state L with age-dependent subjective onset hazard $\lambda_c(a_t)$,

$$\mathcal{Q}_c(L \mid F_0, a_t) = \lambda_c(a_t), \qquad \mathcal{Q}_c(F_0 \mid F_0, a_t) = 1 - \lambda_c(a_t). \tag{22}$$

Conditional on onset, all cohorts face the same full-payment-restoration hazard q_d associated with expected reduced-payment duration d,

$$\mathcal{Q}(F_1 \mid L) = q_d, \qquad \mathcal{Q}(L \mid L) = 1 - q_d. \tag{23}$$

Once payments are restored, the state F_1 is absorbing.

The transition occurs after the current year's choices, so an event following year t first reduces payments in year $t+1$. Households cannot condition current resources on a reduction that has not occurred, but subjective onset risk can change their ex-ante choices through the continuation value. The matrix $\mathcal{Q}_c$ describes what cohort c believes will happen. A second matrix, $\mathcal{Q}^*$, describes what actually happens. Households choose saving, work, and claiming using $\mathcal{Q}_c$. The simulation then draws the aggregate payment history from $\mathcal{Q}^*$, and I measure welfare along that history. In the experiments, $\mathcal{Q}^*$ has depletion at the end of 2033 with probability one, so payments fall in 2034 for every cohort, whereas each $\mathcal{Q}_c$ spreads the depletion date over several years. The two matrices share the same restoration hazard q_d and the same absorbing state F_1. Because beliefs can differ from what actually happens, the matched environments separate outcomes under anticipation from outcomes under surprise while holding the objective payment history fixed.

3.5. Taxes, transfers, assets, and bequests

Households face the 2004 federal tax system. Let y_t^s denote spousal earnings and define other gross income $Y_t^o = \max\{rA_t, 0\} + y_t + y_t^s$, the sum of positive interest income and household labor earnings. The income tax uses the 2004 federal schedule for a head of household with a \$3,100 personal exemption and a \$7,150 standard deduction. Payroll taxes consist of a 6.2-percent OASI tax on earnings up to \$87,900 and an uncapped 1.45-percent Medicare tax. State income taxes are omitted. Social Security benefits are partially taxable, and the statutory rules determine the taxable amount in three steps. The first step computes provisional income, which adds half of the benefit payment to other gross income,

$$\widetilde{Y}_t = Y_t^o + \tfrac{1}{2} ssb_t. \tag{24}$$

Provisional income is the statutory measure that decides whether any benefits are taxed at all. The second step computes a tentative taxable amount from the two statutory thresholds of \$25,000 and \$34,000,

$$T_t^{ss} = \tfrac{1}{2}\min\{ssb_t,\ \widetilde{Y}_t - 25{,}000,\ 9{,}000\} + 0.85\max\{0,\ \widetilde{Y}_t - 34{,}000\}. \tag{25}$$

The first term phases taxation in at a 50-percent rate. It taxes half of the smallest of three quantities, the benefit itself, the excess of provisional income over the \$25,000 threshold, and \$9,000, which is the distance between the two thresholds. The second term adds 85 cents of taxable benefits for every dollar of provisional income above \$34,000. The final step applies the statutory cap,

$$ssb_t^{tax} = \begin{cases} 0, & \widetilde{Y}_t \le 25{,}000, \\ \min\{0.85\, ssb_t,\ T_t^{ss}\}, & \widetilde{Y}_t > 25{,}000. \end{cases} \tag{26}$$

A household below the lower threshold pays no tax on benefits. Above it, the taxable amount is the tentative amount capped at 85 percent of the benefit, so at most 85 percent of benefits are ever taxable. Writing $\mathcal{T}$ for the piecewise-linear after-tax-income function in Appendix C.1, net disposable income is

$$\Upsilon_t = \mathcal{T}(Y_t^o + ssb_t^{tax}) + ssb_t - ssb_t^{tax}. \tag{27}$$

Because ssb_t enters Equation (26), reduced OASI payments also lower the amount subject to benefit taxation. Taxes fall with the payment, partially offsetting the after-tax loss.

Assets evolve according to

$$A_{t+1} = A_t + \Upsilon_t - c_t + tr_t, \qquad A_{t+1} \ge \underline{A}_{t+1}, \tag{28}$$

where the government transfer tr_t implements a consumption floor $\underline{c}$ following Hubbard, Skinner, and Zeldes (1995) through

$$tr_t = \max\Big\{0, \underline{c} - \Big[A_t + \Upsilon_t - \underline{A}_{t+1}\Big]\Big\}. \tag{29}$$

The floor represents means-tested social insurance outside Social Security. It matters for the distributional results. For households with little wealth and low scheduled payments, the floor truncates the downside of the payment shortfall, so their measured welfare effect reflects the safety net as well as the reduction in actual payments. The age-dependent borrowing limit $\underline{A}_{t+1}$ is the natural limit, the largest debt repayable with certainty by the terminal date.

At the end of the terminal period the household values remaining wealth through a warm-glow bequest function following De Nardi (2004),

$$B(A_{T+1}) = \frac{b_1}{1-\eta}(b_2 + A_{T+1})^{1-\eta}, \tag{30}$$

where b_1 scales the strength of the motive and b_2 makes bequests a luxury. The horizon is a fixed terminal age with no additional survival process.

3.6. Recursive formulation

The permanent type indexes separate household problems, and I suppress its notation. The individual state at the beginning of a period, before the leisure shock is realized, is

$$\boldsymbol{X}_t = (A_t, H_t, R_t, s_t, M_t, a_t), \tag{31}$$

collecting assets, human capital, the Social Security record, receipt status, family status, and age. Neither the aggregate payment regime nor the belief group enlarges this individual state vector. The common regime indexes three value functions, and the permanent belief group indexes separate solutions under its own onset hazard. To keep the notation light, I suppress the belief index below. Superscripts F_0, L, and F_1 distinguish the three payment regimes, and $\lambda(a_t)$ denotes the subjective onset hazard for the group being solved.

For $p \in \{0, 1\}$, let $W_t^p(\boldsymbol{X}_t)$ denote the value of branch p before the leisure shock and excluding the current leisure payoff, with $W_t^1 \equiv W_t^W$ and $W_t^0 \equiv W_t^N$. It solves

$$W_t^p(\boldsymbol{X}_t) = \max_{c_t, I_t, d_t} \left\{ \psi_t(M_t)\frac{c_t^{1-\eta}}{1-\eta} + \nu_t d_t + \beta\,\mathbb{E}\Big[\overline{V}_{t+1}(\boldsymbol{X}_{t+1}) \,\Big|\, \boldsymbol{X}_t,\, p_t = p\Big] \right\}, \tag{32}$$

subject to the training set (9), the human-capital technology (12), the Social Security rules, the budget constraint (28), and $I_t = 0$ when $p = 0$. Current OASI payments and disposable resources are evaluated under the current payment regime, whose index is suppressed in W_t^p. Within each branch, consumption and (when working) investment are continuous choices and claiming is discrete. The expectation is taken over next-period family status, spousal earnings, and the human-capital innovation. Appendix D writes out this operator and the cases in which its integrals degenerate.

The current payment state affects current OASI payments, taxable benefits, disposable resources, and the budget constraint through equations (20)–(28). Transition probabilities enter the continuation value $\overline{V}_{t+1}$, the household's expected value of next period given the current payment state z_t. There are three cases. If payments have already been restored ($z_t = F_1$), the state never changes again, so the continuation is simply $V^{F_1}_{t+1}$. If payments are currently reduced ($z_t = L$), the household expects restoration next year with probability q_d and continued reduced payments with probability $1 - q_d$, so the continuation is a weighted average of $V^{F_1}_{t+1}$ and V^{L}_{t+1}. If payments are still full and depletion has not yet occurred ($z_t = F_0$), the household expects depletion next year with subjective probability $\lambda(a_t)$ and continued full payments with probability $1 - \lambda(a_t)$, so the continuation is a weighted average of V^{L}_{t+1} and $V^{F_0}_{t+1}$. Formally,

$$\overline{V}_{t+1}(\boldsymbol{X}_{t+1}) = \begin{cases} V^{F_1}_{t+1}(\boldsymbol{X}_{t+1}), & z_t = F_1, \\ [1 - q_d]V^{L}_{t+1}(\boldsymbol{X}_{t+1}) + q_d V^{F_1}_{t+1}(\boldsymbol{X}_{t+1}), & z_t = L, \\ [1 - \lambda(a_t)]V^{F_0}_{t+1}(\boldsymbol{X}_{t+1}) + \lambda(a_t)V^{L}_{t+1}(\boldsymbol{X}_{t+1}), & z_t = F_0. \end{cases} \tag{33}$$

Finally, the value at the start of the period, before the leisure shock is drawn, is the expected maximum of the two work branches. Using the closed form in Equation (8),

$$V_t(\boldsymbol{X}_t) = \mathcal{E}_t\Big(W^W_t(\boldsymbol{X}_t),\, W^N_t(\boldsymbol{X}_t)\Big)\,. \tag{34}$$

The structure of Equation (33) dictates the solution order. The restored full-payment problem V^{F_1} is solved first. The reduced-payment problem V^L is then solved using V^{F_1} and the restoration hazard q_d. Finally, the pre-depletion full-payment problem V^{F_0} is solved once per belief group using V^L and that group's subjective onset hazard.

3.7. Solution method

I solve the finite-horizon model by backward induction. At each age and state, I compare the admissible work and claiming branches, optimize consumption and human-capital investment within each branch, and integrate the next-period shocks. Appendix D reports the grids, numerical integration, interpolation, optimization, and Social Security calculation order.

3.8. Parameterization and benchmark fit

I retain the parameter estimates of Fan, Seshadri, and Taber (2024) to preserve their parsimonious life-cycle benchmark and isolate the payment-regime extension. Figure 4 compares the inherited benchmark with the employment, wage, consumption, and Social Security claiming profiles used in their analysis. The fit is closer for some moments than for others, so I use the figure as a transparent benchmark check rather than as evidence of uniform fit. Setting perceived onset risk to zero and $\chi = 1$ returns the full-payment benchmark.

The extension adds the common objective reserve-depletion event, the Trustees-informed subjective onset hazards described in Section 4, and the full-payment-restoration hazards. These calendar-time objects map into different age profiles across birth cohorts while preferences and the remaining model primitives stay fixed. The empirical hazards in Section 2.2 provide an alternative belief specification.

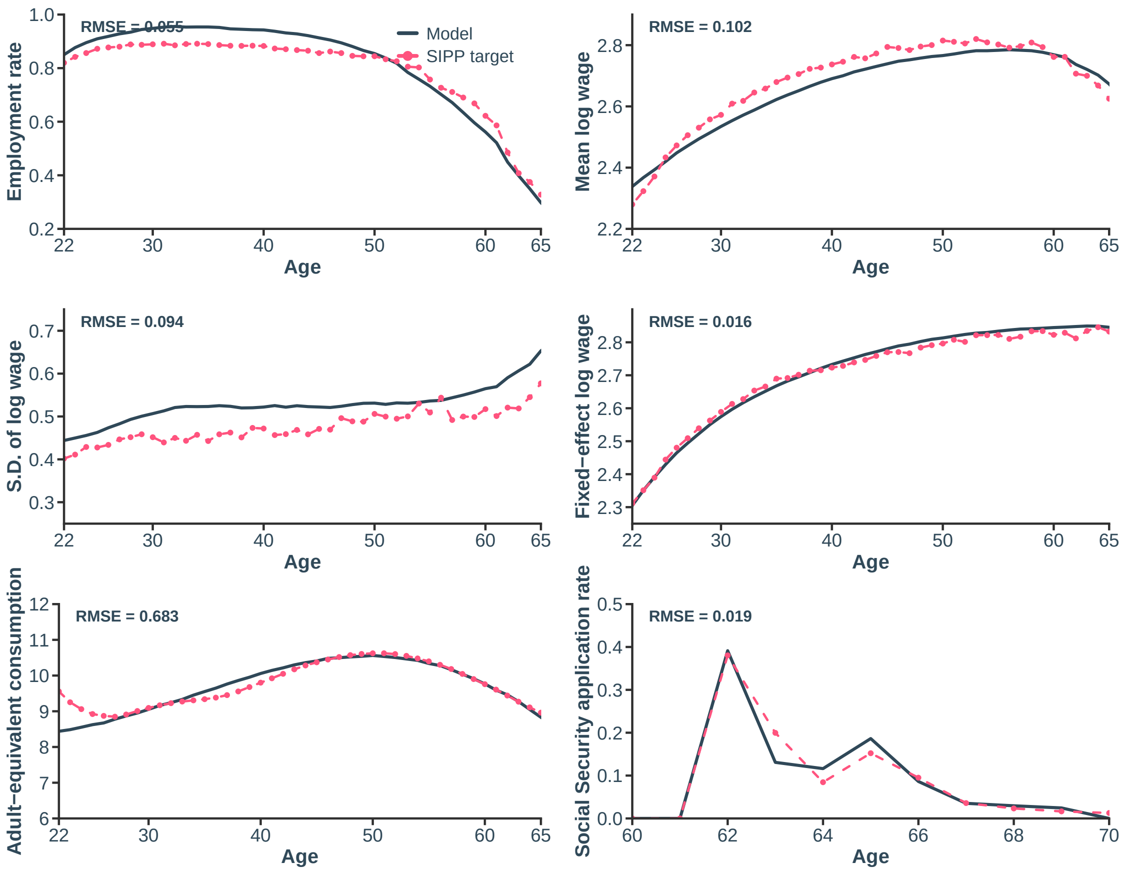

FIGURE 4. Replication of Fan, Seshadri, and Taber (2024) benchmark

Note. The figure compares the Fan, Seshadri, and Taber (2024) benchmark with the corresponding empirical target profiles. Household parameters are inherited from their work and are not re-estimated using the payment-shortfall experiments.

4. Counterfactuals and Welfare Analysis

The analysis distinguishes two conceptually different comparisons. The first is the surprise-shortfall effect: a household that unexpectedly receives 78 percent of its scheduled OASI payment reoptimizes consumption, saving, work, human-capital investment, and claiming after reduced payments begin. The second is the anticipation effect: a household that assigns positive probability to payment-shortfall onset can adjust those choices before reserve depletion.

Anticipatory adjustments can reduce the marginal utility cost of reduced payments, but they are themselves costly because they reallocate consumption and leisure across ages. A single comparison conflates these two components. I therefore simulate the same households, with identical preference parameters, initial conditions, idiosyncratic shocks, and objective payment histories, under the three matched environments in Table 1.

TABLE 1. Counterfactual information and payment histories

Environment	Information used for choices	Objective payment history
Full-payment benchmark (O)	No perceived onset risk	Full scheduled payments
Surprise shortfall (U)	Full-payment policy through the 2033 event	Reduced payments begin in 2034; restoration hazard q_d
Anticipated shortfall (A)	Cohort-specific onset beliefs and restoration hazard q_d	The same distribution as the surprise environment

From a calendar distribution to subjective onset hazards. Two belief objects must remain distinct. The first is a common subjective distribution over the calendar year of OASI reserve depletion. The second is a cohort-specific subjective onset hazard, which expresses that same calendar distribution in the age of a particular birth cohort. The calendar distribution is primitive; the age-specific hazard is derived.

Calendar distribution. Let D denote the end-of-year date of OASI reserve depletion, and let $\pi(y) \equiv \Pr(D = y \mid D \geq 2027)$ denote its subjective probability conditional on reserves remaining positive through the 2026 base year. The distribution $\{\pi(y)\}$ is defined over calendar years and is common across birth cohorts. The counterfactuals use a Trustees-informed distribution centered on 2032, with essentially all mass between 2030 and 2037.[5]

Cohort-specific hazard. The model is solved in age time. For a household born in year b, calendar year y arrives at age $a = y - b$. The subjective onset hazard at age a is therefore the probability that depletion falls in calendar year $b + a$, divided by the probability that it has not occurred earlier,

$$\lambda_b(a) = \frac{\pi(b+a)}{\sum_{y \geq b+a} \pi(y)}. \tag{35}$$

Equation (35) is a change of variables: every cohort's hazard vector $\{\lambda_b(a)\}_{a=18}^{80}$ is generated from the same $\{\pi(y)\}$. Two households born in different years therefore hold identical beliefs about the calendar depletion date but face that risk at different ages. For example, the conditional probability of depletion at the end of 2032 appears as $\lambda_{1957}(75)$ for the 1957 cohort and $\lambda_{2005}(27)$ for the 2005 cohort. The remaining adjustment horizons differ by 48 years. Because end-of-year depletion first lowers payments in the following year, the hazard enters the continuation value one period before payments can change.

Two sources of uncertainty must be distinguished: when reduced payments begin and how long they last. Onset beliefs are governed by $\lambda_b(a)$, whereas the common objective depletion date is fixed at the end of 2033. Duration is governed by the full-payment-restoration hazard $q_d = 1/d$ for finite d and $q_d = 0$ under permanence. This hazard is common to household beliefs and the objective payment process. Differences between the surprise and anticipated environments therefore reflect anticipation of onset, not different beliefs about persistence.

The three environments. The full-payment benchmark (O) is the reference environment. Households assign zero probability to payment-shortfall onset and receive full scheduled OASI payments at every age. This payment path is imposed exogenously. The model does not specify how full scheduled payments would be financed and does not charge households

[5] Two report vintages play different roles. The fixed end-of-2033 objective event follows the 2025 Trustees Report's OASI point projection. The subjective onset distribution uses the 2026 update, which moved the OASI point projection to the fourth quarter of 2032. To translate the 2026 stochastic information into an OASI distribution, I shift the reported combined-OASDI range to align with the OASI point projection, fit a split-normal distribution with median 2032 and 2.5/97.5 percentiles of 2030/2037, discretize it by year, and condition on reserves remaining positive through 2026. Thus, the 2033 event is a controlled design input, while $\pi(y)$ describes the beliefs used for ex-ante choices. Section 2.2 provides an alternative mapping from measured survey beliefs.

for that financing; the benchmark is therefore a conditional incidence comparison, not a complete welfare evaluation of a reform.

The surprise-shortfall environment (U) imposes the same objective payment history as the anticipated environment but removes foresight. Through the end of 2033, households follow the full-payment policy. At the start of 2034, they learn that the paid-benefit fraction is $\phi = 0.78$ and that full-payment restoration arrives with hazard q_d. They then reoptimize under the reduced-payment policy V^L. This environment contains ex-post behavioral adjustment but no anticipatory adjustment.

In the anticipated-shortfall environment (A), cohort c uses the subjective onset-hazard vector $\lambda_c(a)$ and restoration hazard q_d in every decision from labor-market entry onward. Its objective payment histories are identical to those in U. Any difference between A and U therefore arises from choices made under anticipation rather than from different realized payments.

Decomposition. Let Y_{cd} denote any outcome for cohort c under expected duration d, and let Y_{cd}^O, Y_{cd}^U, and Y_{cd}^A denote its value under the three environments. Pairwise differences define three effects:

$$\Delta_{cd}^U = Y_{cd}^U - Y_{cd}^O, \qquad \text{surprise-shortfall effect,} \tag{36}$$

$$\Delta_{cd}^P = Y_{cd}^A - Y_{cd}^U, \qquad \text{anticipation effect,} \tag{37}$$

$$\Delta_{cd}^T = Y_{cd}^A - Y_{cd}^O, \qquad \text{total anticipated-shortfall effect.} \tag{38}$$

The surprise-shortfall effect Δ_{cd}^U combines the loss of OASI payment income with the ex-post behavioral adjustment made after reduced payments begin. It is not a mechanical effect because households reoptimize after the surprise. The fixed-behavior exposure measure in Section 5.3 separately identifies the mechanical timing channel.

The anticipation effect Δ_{cd}^P compares A with U, holding the objective payment histories fixed. Its sign is not determined a priori. Anticipatory choices can sustain consumption during the reduced-payment spell, but they also require earlier consumption and leisure sacrifices. This comparison is not a pure value-of-information measure because subjective onset beliefs need not match the objective depletion date.

The total anticipated-shortfall effect Δ_{cd}^T compares the anticipated-shortfall environment directly with the full-payment benchmark and is the primary object for intergenerational welfare incidence. The underlying utility gaps telescope across the three environments, but each CEV is a separate nonlinear transformation and is computed from its own target–reference comparison. Section 5 reports all three comparisons for each birth cohort and expected reduced-payment duration.

4.1. Welfare Objects

The welfare object is a consumption-equivalent variation (CEV), which converts a lifetime utility difference into a percentage of the reference environment's consumption path. Each comparison in equations (36)– (38) pairs a target environment j with a reference environment k. The CEV chooses the common adjustment to reference consumption that closes the utility difference,

$$U_c^k(\lambda_c) = U_c^j(1), \qquad \text{CEV} = 100\,(1 - \lambda_c). \tag{39}$$

Specifically, I scale consumption in the reference environment by a common factor $\lambda_c > 0$ at every age from 18 through 80, leaving its other simulated choices fixed. Cohort-specific lifetime utility under this adjustment is

$$U_c^k(\lambda_c) = \mathbb{E}\left[\sum_{a=18}^{80} \beta^{a-18}\, u_a\Big(\lambda_c\, c_{ca}^k,\, p_{ca}^k,\, d_{ca}^k\Big) + \beta^{63} B\Big(A_{c,81}^k\Big)\right],$$

for $k \in \{O, U, A\}$, where u_a is flow utility from equation (4) and B is the warm-glow bequest from equation (30). For the utility function in equation (4), define the reference environment's expected discounted consumption-utility component as

$$\mathcal{C}_c^k = \mathbb{E}\left[\sum_{a=18}^{80} \beta^{a-18} \psi_a(M_a) \frac{(c_{ca}^k)^{1-\eta}}{1-\eta}\right], \qquad \mathcal{N}_c^k = U_c^k(1) - \mathcal{C}_c^k.$$

Because only reference consumption is rescaled, $U_c^k(\lambda) = \lambda^{1-\eta}\mathcal{C}_c^k + \mathcal{N}_c^k$. Equating this expression with target utility gives

$$\lambda_c = \left(\frac{U_c^j(1) - \mathcal{N}_c^k}{\mathcal{C}_c^k}\right)^{\frac{1}{1-\eta}}. \tag{40}$$

The discount factor β and family-size shifters $\psi_a(M_a)$ remain inside $\mathcal{C}_c^k$.

Two points clarify interpretation. First, the CEV rescales a fixed reference consumption allocation while leaving labor supply, claiming, and wealth at their simulated values. It is a unit conversion, not a further behavioral reoptimization. Second, λ_c matches cohort-average utility, so the measure is a cohort-level welfare effect rather than a household-level statistic.

Under equation (39), a positive CEV denotes a welfare cost: $\lambda_c < 1$ when the target environment has lower lifetime utility than the reference. For example, comparing the anticipated-shortfall environment with the full-payment benchmark for the 1982 cohort in the canonical simulation gives $\lambda_{1982} = 0.987385$, or a welfare cost of 1.261 percent. This means that a 1.261-percent reduction in benchmark consumption at every age from 18 to 80 would generate the same cohort-average lifetime utility as the target environment.

5. Intergenerational Welfare Incidence

This section asks how expected reduced-payment duration changes welfare incidence across birth cohorts. I report five cohorts—1957, 1965, 1975, 1982, and 2005—at ages 76, 68, 58, 51, and 28, respectively, at the common end-of-2033 OASI reserve-depletion date. These ages place the event after, near, and well before the retirement transition.

The permanent reduced-payment regime is one endpoint, not the definition of an OASI payment shortfall. I therefore allow full scheduled payments to resume and vary the expected duration of the reduced-payment state. Longer expected durations increasingly move the welfare burden toward cohorts whose retirement lies farther in the future.

The exercise separates two mechanisms: how persistence shifts mechanical benefit exposure from older to younger cohorts, and how anticipatory behavior changes the associated consumption-equivalent welfare effects.

5.1. Varying Expected Reduced-Payment Duration

I change one object: the expected duration of the reduced-payment state. The common objective depletion date remains at the end of 2033, reduced payments begin in 2034, and the paid-benefit fraction remains $\phi = 0.78$. Statutory entitlements and other program rules do not change. Full-payment restoration ends the spell; unpaid scheduled payments from earlier years are never repaid. The taxes or spending changes that finance restoration are unspecified and excluded from household welfare.

Duration is parameterized by a constant restoration hazard. For each expected duration $d \in \{1, 2, 3, 5, 10, 20, 40\}$, the reduced-payment state ends with annual hazard $q_d = 1/d$. Conditional on being in state L,

$$\Pr(F_{1,t+1} \mid L_t) = q_d, \qquad \Pr(L_{t+1} \mid L_t) = 1 - q_d, \qquad q_d = \frac{1}{d}, \tag{41}$$

where F_1 is the restored full-payment state. The realized reduced-payment duration D is geometrically distributed with mean d. Thus d denotes an expected duration, not a predetermined number of reduced-payment years. When $d = 1$, full scheduled payments resume with certainty after one year. When $d = 5$, they resume after each reduced-payment year with probability 0.2, so the spell lasts five years on average but may end sooner or later.

For every duration, the model solves the three environments in equations (36)–(38), denoted Y^O_{cd}, Y^U_{cd}, and Y^A_{cd}. The full-payment benchmark receives full scheduled payments throughout life. In the surprise-shortfall environment, households follow the benchmark policy through 2033 and reoptimize when reduced payments begin. In the anticipated-shortfall environment, households use cohort-specific subjective onset hazards and anticipate the same restoration process.

I denote the corresponding outcome differences by Δ^U_{cd} for the surprise-shortfall effect, Δ^P_{cd} for the anticipation effect, and Δ^T_{cd} for the total anticipated-shortfall effect. Their consumption-equivalent measures are $\mathcal{C}^U_{cd}$, $\mathcal{C}^P_{cd}$, and $\mathcal{C}^T_{cd}$. A positive value denotes a welfare cost relative to the named reference environment.

Computation follows one canonical protocol. Each cohort–duration cell contains 20,000 simulated people, and common random numbers hold idiosyncratic histories fixed across the three matched environments. For stochastic restoration, lifetime utility is first integrated over the common distribution of restoration histories, then averaged across people, and finally transformed into the cohort-level CEV in equation (40). Every structural table and figure that reports the permanent simulation uses this same endpoint. Reported intervals are paired conditional Monte Carlo intervals. They quantify finite-simulation variation with model parameters and empirical inputs held fixed; they do not incorporate parameter, model, empirical, or multiple-comparison uncertainty.

5.2. The duration frontier

Figure 5 is the main result. It plots the consumption-equivalent welfare cost against expected reduced-payment duration. Panel B reports the surprise-shortfall cost, $\mathcal{C}^{U}_{cd}$, which compares the surprise-shortfall environment with the full-payment benchmark. Households do not adjust before 2034 in either environment, but they reoptimize after the surprise; this comparison therefore measures the model-implied consequence of the payment shortfall, not mechanical exposure alone.

The surprise-shortfall cost is non-decreasing in expected duration for every cohort, but its rate of increase differs sharply across cohorts. Older cohorts are affected even by short episodes, but they have relatively few remaining benefit years. For the 1957 cohort, which is age 77 when reduced payments begin, the surprise-shortfall cost rises from 0.058 percent under a one-year shortfall to 0.229 percent under permanence. For the 1965 cohort, it rises from 0.069 to 0.756 percent.

Younger cohorts exhibit the opposite pattern. The 1975, 1982, and 2005 cohorts bear essentially no surprise-shortfall cost under a one-year spell because they receive little or no OASI before payments are restored. Their costs become substantial only when reduced payments persist into their retirement years. At an expected duration of ten years, the surprise-shortfall costs are 0.345 percent for the 1975 cohort, 0.182 percent for the 1982 cohort, and 0.036 percent for the 2005 cohort. Under permanent reduced payments, they rise to 1.157, 1.194, and 1.334 percent, respectively. The ranking reverses: among these three cohorts, the oldest bears the largest cost at ten years, whereas the youngest bears the largest cost under permanence. This reversal illustrates how greater persistence shifts the welfare burden toward younger generations.

Panel A reports the total CEV from the direct comparison of the anticipated shortfall with the full-payment benchmark. At an expected duration of one year, the 1965 cohort has the largest total cost. It remains the most affected cohort at five years, with a cost of 0.262 percent, compared with 0.137 percent for the 1957 cohort and 0.122 percent for the 1975 cohort. By twenty years, the point estimate for the 1975 cohort exceeds that for the 1965 cohort: 0.668 versus 0.619 percent. Under permanent reduced payments, each of the three youngest cohorts bears a consumption-equivalent welfare cost of approximately 1.26 percent, compared with 0.75 percent for the 1965 cohort and 0.18 percent for the 1957 cohort. Thus, as the reduced-payment state becomes more persistent, the largest welfare

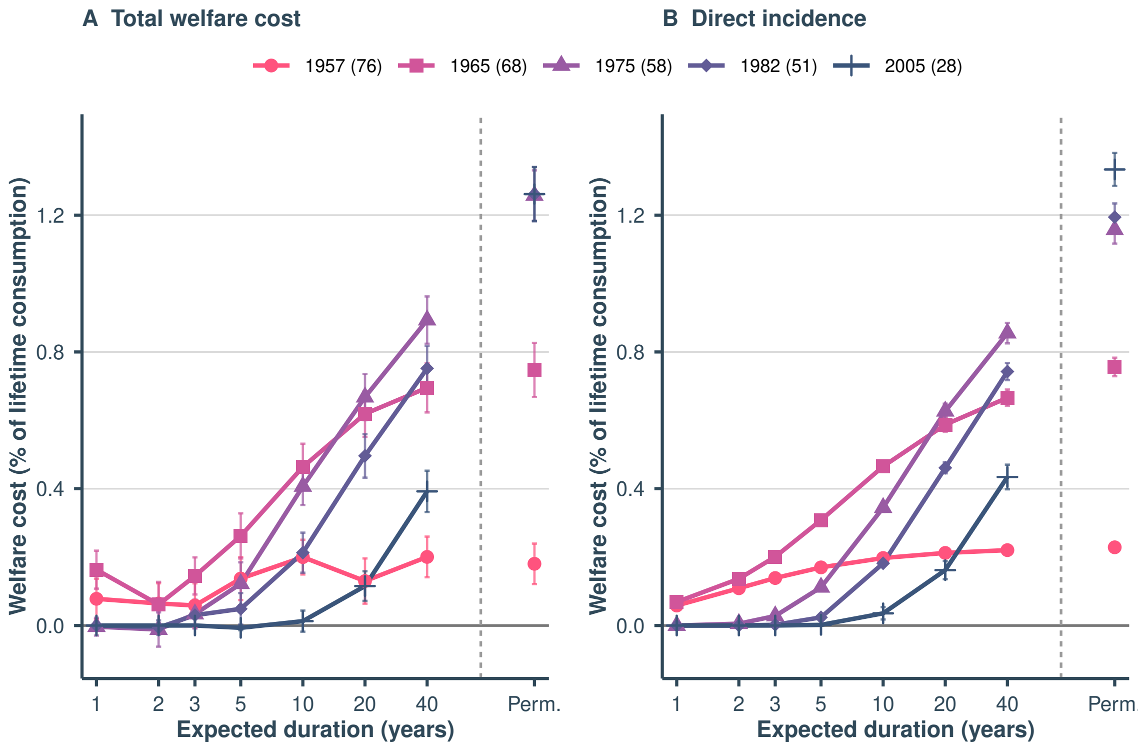

FIGURE 5. Expected reduced-payment duration and intergenerational welfare costs

Note. OASI reserves are depleted at the end of 2033, and reduced payments begin in 2034 at a paid-benefit fraction of 0.78. The horizontal axis gives expected reduced-payment duration; "Permanent" denotes zero restoration hazard. Panel A compares the anticipated-shortfall environment with the full-payment benchmark. Panel B compares the surprise-shortfall environment with the full-payment benchmark and allows ex-post reoptimization after 2034. Welfare costs are consumption-equivalent variations, with losses shown as positive values. Vertical bars are paired 95-percent conditional Monte Carlo intervals from 20,000 simulated people per cohort–duration cell. They hold parameters and empirical inputs fixed and do not cover model, estimation, or multiple-comparison uncertainty.

burden shifts from cohorts near retirement toward younger generations. The permanent values reported here are also used in Table E.1 and the appendix life-cycle figures; there is no second permanent simulation with different draws.

Total welfare costs do not, however, rise mechanically with expected duration. The surprise-shortfall cost increases with duration, but the anticipation effect can change sign, causing the total cost frontier to be non-monotonic. The 1965 cohort provides the clearest example. When expected duration increases from one to two years, its surprise-shortfall cost rises from 0.069 to 0.137 percent. Over the same comparison, the anticipation effect changes from a cost of 0.095 percentage points to a cost reduction of 0.076 percentage points. The total welfare cost consequently falls from 0.163 to 0.061 percent despite the increase in objective exposure.

The small negative total-cost point estimates for some younger cohorts at expected durations of two to five years should not be interpreted as evidence that reduced payments improve welfare. These cohorts have negligible mechanical benefit exposure at short durations, so their point estimates are driven primarily by the estimated anticipation effect. The paired conditional Monte Carlo intervals reflect only finite-simulation variation under common random numbers. They do not account for uncertainty in estimated parameters, empirical inputs, or the model itself.

5.3. Benefit Exposure

The preceding subsection shows that longer reduced-payment spells shift the surprise-shortfall welfare burden toward younger cohorts. That pattern alone does not identify the mechanism because households reoptimize after the surprise. This subsection isolates the fixed-behavior timing channel.

For each cohort, I first calculate how its discounted lifetime Social Security benefits are distributed across ages. Let s_{ca} denote the share of cohort c's discounted lifetime scheduled payments received at age a, measured along the full-payment benchmark path. These shares sum to one. Let p_{cda} denote the probability that the reduced-payment state is still in effect when cohort c reaches age a, given expected duration d. The cohort's mechanical benefit exposure is

$$E_{cd} = \sum_{a} s_{ca} p_{cda}. \tag{42}$$

This is a weighted average of the probabilities of reduced payments, where ages accounting for a larger share of the cohort's Social Security wealth receive greater weight. A value of zero means that none of the cohort's scheduled payments are expected to fall during the reduced-payment state, while a value of one means that its entire benefit stream is exposed. Because the benefit shares s_{ca} are calculated from the full-payment benchmark path, differences in E_{cd} reflect only the timing and persistence of the reduced-payment state, not changes in claiming, earnings, or other household behavior.

Only the probability of being in the reduced-payment state, p_{cda}, varies across cohorts and durations. The resulting exposure measure captures the mechanical effect of life-cycle timing rather than households' behavioral responses. Table 2 reports the exposure measure at three durations. It also reports the probability that each cohort ever receives a reduced

TABLE 2. Mechanical benefit exposure across cohorts and durations

	Exposure share E_{cd} (percent)			Pr(ever reduced)	Expected reduced years
Cohort (age in 2033)	$d = 1$	$d = 20$	Permanent	$d = 20$	Permanent
1957 (76)	5.1	18.3	19.7	1.000	4
1965 (68)	6.2	51.9	66.7	1.000	12
1975 (58)	0.0	55.1	100.0	0.785	17.17
1982 (51)	0.0	38.5	100.0	0.548	17.17
2005 (28)	0.0	11.8	100.0	0.168	17.17

Note. Exposure is the discounted share of scheduled OASI payments in the full-payment benchmark that falls in reduced-payment years, equation (42). Permanent denotes the zero-restoration-hazard boundary. Pr(ever reduced) is the probability of ever receiving a reduced payment at an expected duration of twenty years; expected reduced years are expected years of reduced-benefit receipt under permanence. Entries are computed from the full-payment benchmark's simulated scheduled-payment paths.

payment and the expected number of years for which it receives reduced benefits. The table shows how the affected population changes as the reduced-payment state becomes more persistent. Under a one-year spell, only the two oldest cohorts receive reduced payments. Their exposure shares are 5.1 percent for the 1957 cohort and 6.2 percent for the 1965 cohort. The three younger cohorts have zero exposure because full payments are restored before they begin receiving OASI.

At an expected duration of twenty years, reduced payments reach cohorts that are closer to retirement. Exposure is then hump-shaped across cohorts, reaching 55.1 percent for the 1975 cohort and 38.5 percent for the 1982 cohort. Their probabilities of ever receiving a reduced payment are 0.785 and 0.548, respectively. The 2005 cohort remains much less exposed: its probability of receiving a reduced payment is 0.168, and its exposure share is 11.8 percent. Thus, a moderately persistent shortfall primarily affects cohorts approaching claiming eligibility; only a longer shortfall reaches workers who are farther from retirement.

Two life-cycle features determine these patterns. Older cohorts are hit even by a short reduced-payment spell, but only a small share of their lifetime benefits is still ahead of them. Younger cohorts avoid short episodes because they have not yet begun receiving benefits, but they have a much larger remaining benefit stream at risk when reduced payments persist. Under permanent reduced payments, 19.7 percent of the 1957 cohort's discounted lifetime benefits and 66.7 percent of the 1965 cohort's benefits are exposed. By contrast, all scheduled benefits of the 1975, 1982, and 2005 cohorts are exposed. These cohorts receive reduced benefits for an expected 17.17 years, compared with 12 years for

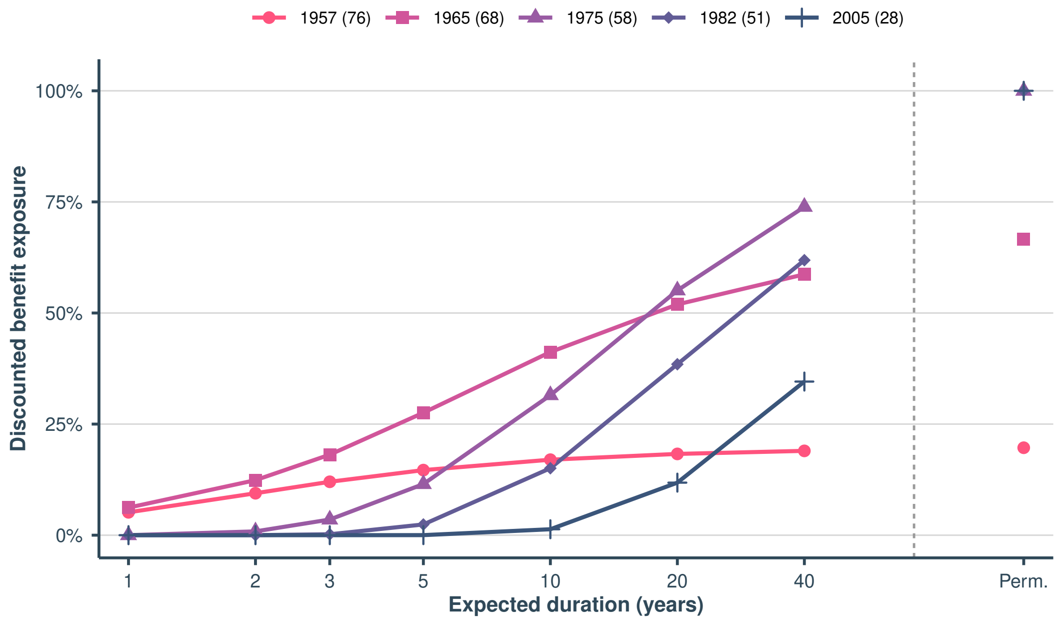

FIGURE 6. Mechanical benefit exposure across cohorts and durations

Note. The figure reports the discounted share of each cohort's scheduled OASI benefits expected to be received while payments are reduced. Benefits are evaluated along the full-payment benchmark path, so differences across cohorts and durations reflect life-cycle timing and the persistence of the reduced-payment state rather than behavioral responses. Lines connect the seven modeled finite-duration scenarios as visual guides. Permanent denotes the zero-restoration-hazard case and is plotted separately. Legend labels report birth cohort, with age in 2033 in parentheses.

the 1965 cohort and 4 years for the 1957 cohort.

Figure 6 summarizes how this mechanical exposure changes with duration. Short episodes primarily affect cohorts already receiving benefits, whereas longer episodes increasingly expose younger cohorts. The curves provide the life-cycle explanation for the nonlinear welfare incidence documented above: as reduced payments persist, a larger share of younger cohorts' retirement benefits falls within the shortfall period.

A Stylized Example. The mechanism above can be reproduced without the structural model's simulated earnings, claiming decisions, or policy functions. To show this, I replace each household with a normalized scheduled OASI annuity that pays one unit from age 67 through age 110. The annuity is discounted to 2026 at $\beta = 0.97$. OASI reserves are depleted at the end of 2033, reduced checks begin in 2034, and each reduced check equals 78 percent of its scheduled amount. The only uncertainty is how long reduced payments persist. If m_h is the ratio of paid to scheduled discounted annuity wealth in policy h, the

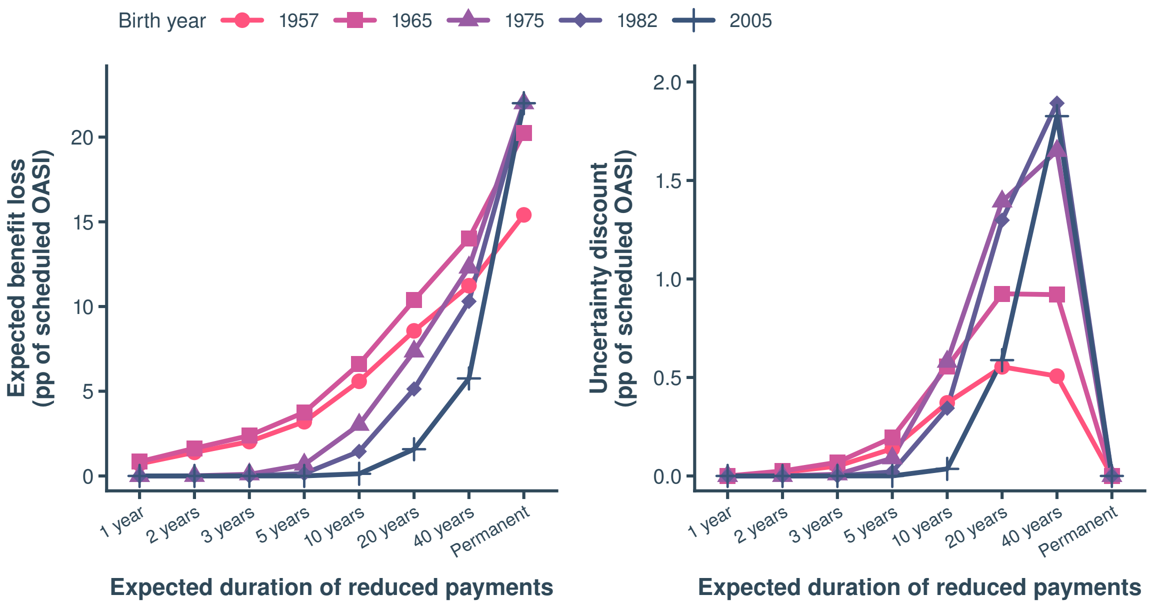

FIGURE 7. A fixed-behavior benchmark for benefit loss and uncertainty

Note. The scheduled annuity pays one unit from age 67 through age 110 and is discounted to 2026 at 0.97. OASI reserves are depleted at the end of 2033; reduced checks begin in 2034 and pay 78 percent of their scheduled amount. Finite durations are geometric restoration processes indexed by their expected duration. Panel A reports $100(1 - \mathbb{E}[m])$; Panel B reports $100(\mathbb{E}[m] - CE(m))$, with CRRA coefficient 4.0363. One-year duration and permanence are deterministic conditional on onset. All histories are enumerated exactly.

calculation reports[6]

$$1 - CE(m) = \underbrace{1 - \mathbb{E}[m]}_{\text{expected accounting loss}} + \underbrace{\mathbb{E}[m] - CE(m)}_{\text{uncertainty discount}}, \tag{43}$$

where the certainty equivalent uses the benchmark CRRA coefficient.

This calculation depends only on the scheduled annuity and the distribution of payment histories; it holds behavior fixed and enumerates those histories exactly. The first term is the expected accounting loss. The second is the uncertainty discount induced by dispersion in realized reduced-payment durations. Neither term is a structural life-cycle welfare effect.

Figure 7 reports the fixed-behavior calculation. The longer reduced payments last,

[6]For clarity, the certainty equivalent is computed as

$$CE(m) = u^{-1}(\mathbb{E}[u(m)]) = \begin{cases} \left(\mathbb{E}[m^{1-\gamma}]\right)^{1/(1-\gamma)}, & \gamma \neq 1, \\ \exp\{\mathbb{E}[\log m]\}, & \gamma = 1. \end{cases}$$

the greater the expected loss, especially for younger cohorts whose retirement is still many years away. The uncertainty discount first increases and then decreases. A one-year reduced-payment spell and permanent reduced payments have no duration uncertainty conditional on reserve depletion, so their uncertainty discount is zero. Permanent reduced payments can nevertheless cause a large average loss. The uncertainty discount is largest when expected reduced-payment duration is 20 years for the 1957 and 1965 cohorts and 40 years for the 1975, 1982, and 2005 cohorts. Even with behavior fixed, expected duration matters because cohorts reach the common depletion date at different ages.

The benchmark gives two additional points. First, consider the 1982 cohort and three restoration processes. All have the same expected accounting loss of 4.2 percentage points, but different uncertainty about the restoration date. A known date produces no uncertainty discount. A constant annual restoration hazard produces a discount of 1.1 percentage points. A mixture of quick restoration and permanent reduced payments produces a discount of 2.0 percentage points. Expected loss alone therefore does not summarize duration risk.

Second, I compare uncertainty about the reserve-depletion date with uncertainty about realized reduced-payment duration. I allow reserve depletion to occur between 2027 and 2050 and set expected reduced-payment duration to 20 years. An allocation of the total uncertainty discount assigns more than 98 percent to duration uncertainty for every cohort. This calibration-dependent result supports the paper's focus on persistence after depletion. Appendix C.2 provides the exact calculations.

5.4. The Welfare Effect of Anticipation

Section 5.3 holds behavior fixed. I now compare the anticipated- and surprise-shortfall environments to measure the welfare effect of choices made under subjective onset beliefs. Households can adjust saving, employment, consumption, human-capital investment, and claiming before reduced payments begin. The objective depletion date, paid-benefit fraction, restoration process, and payment histories are identical across the two environments. The anticipation effect has no uniform sign: it lowers the welfare cost for some cohort-duration pairs and raises it for others.

Figure 8 reports $\mathcal{C}^{P}_{cd}$. A positive value means that anticipation raises the consumption-equivalent welfare cost relative to surprise under the same objective payment process; a negative value means that anticipation lowers it. The signs are heterogeneous because

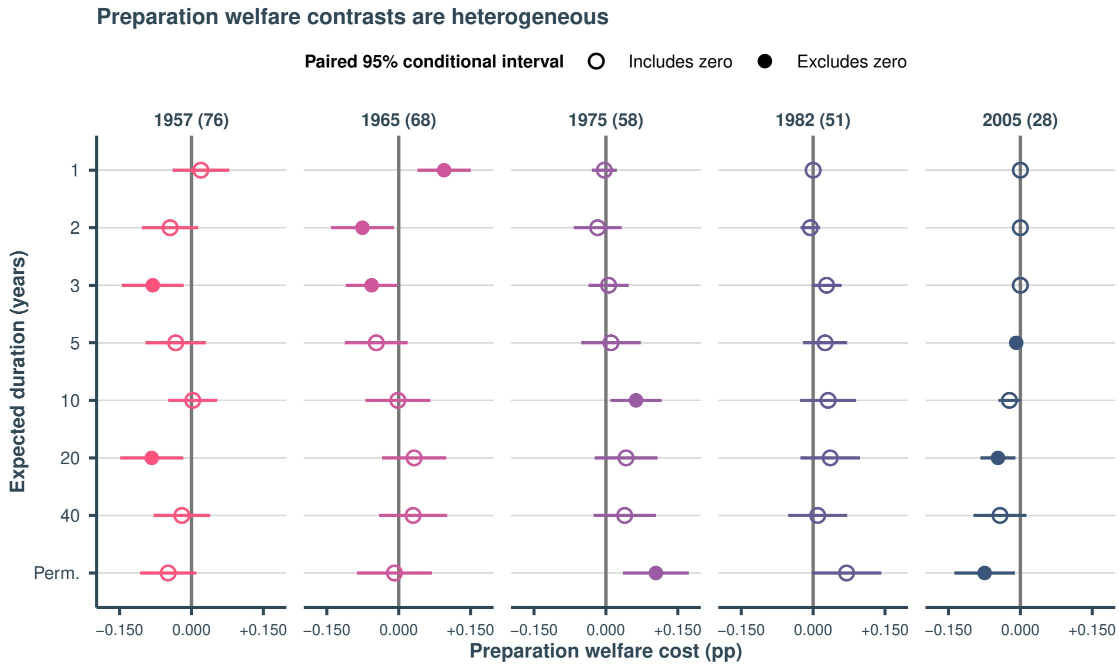

FIGURE 8. Welfare effect of anticipation by cohort and duration

Note. Each point reports $\mathcal{C}^{P}_{cd}$. Positive values mean that anticipation raises the consumption-equivalent welfare cost; negative values mean that it lowers the cost. Filled markers denote paired 95-percent conditional Monte Carlo intervals that exclude zero for contrasts larger than the numerical tolerance of 10^{-10}; open markers denote all other cases. The 2005 contrasts at one and two years are machine zeros of order 10^{-14} and are shown as open markers. The intervals use 20,000 simulated people per cohort–duration cell and cover finite-simulation variation only.

self-insurance has both benefits and costs. Anticipation lets households accumulate assets, adjust work and claiming, and, with enough time, invest in human capital before reduced payments arrive. Those choices can sustain consumption during the reduced-payment spell, but they can also lower earlier consumption and leisure or move resources away from states in which their marginal value is high.

The permanent reduced-payment case makes the differences across cohorts especially clear. For the 1975 cohort, which is age 58 in 2033, anticipating the payment shortfall raises the lifetime welfare cost by 0.104 percentage points relative to being surprised by the same shortfall (95-percent conditional Monte Carlo interval $[0.035, 0.173]$). These households have time to adjust, but doing so requires changes in saving, work, and claiming during a period of life when those choices are already important. In this case, the lifetime cost of those earlier adjustments is larger than their benefit from smoothing consumption after payments fall.

For the 1982 cohort, the estimated effect is also positive, at 0.070 percentage points. However, its 95-percent conditional Monte Carlo interval ($[-0.003, 0.142]$) includes zero. Conditional on the calibrated model and empirical inputs, simulation variation therefore does not separate this contrast from zero. The pattern differs for the 2005 cohort, which is age 28 in 2033. For this group, anticipating permanent reduced payments lowers the welfare cost by 0.075 percentage points (95-percent conditional Monte Carlo interval $[-0.138, -0.011]$). Because these households have many working years before retirement, they can spread adjustments in saving, work, and human-capital investment over a longer part of their lives. Finally, the estimated effects for the 1957 and 1965 cohorts are close to zero, and their conditional intervals include zero.

This pattern is not limited to permanent reduced payments. For the 1965 cohort, anticipation raises the welfare cost by 0.095 percentage points when expected duration is one year, but lowers it by 0.076 percentage points when expected duration is two years. For the 1957 cohort, the conditional intervals exclude zero at expected durations of three and twenty years. For the 2005 cohort, they exclude zero at five years, twenty years, and permanence.

The total welfare cost can therefore rise or fall with expected duration because the anticipation effect can have either sign. Among the 40 cohort–duration comparisons, ten economically nonzero contrasts have paired conditional Monte Carlo intervals that exclude zero. Two additional machine-zero contrasts have degenerate numerical intervals and are treated as zeros. This count is a descriptive simulation diagnostic, not a general significance test: the intervals hold parameters and empirical inputs fixed and are not adjusted for multiple comparisons. The conclusion is not that anticipation is generally beneficial or costly. Its welfare effect depends on life-cycle position and expected reduced-payment duration.

5.5. Accounting Incidence versus Welfare Incidence

Finally, I compare the model-based welfare cost with a fixed-behavior accounting exercise that converts the present value of unpaid scheduled payments into an annuity-equivalent consumption loss. Table 3 reports the comparison; Appendix E.4 discusses each panel in detail.

The comparison delivers two lessons. First, a ranking based on unpaid scheduled

TABLE 3. Accounting incidence versus welfare incidence

	1957	1965	1975	1982	2005
Panel A. Accounting loss: unpaid scheduled OASI payments, behavior fixed					
Share of scheduled OASI wealth lost (%)	4.3	14.7	22.0	22.0	22.0
Loss as % of lifetime consumption	0.29	0.99	1.49	1.49	1.49
Loss in present value at 2033 (peak = 100)	34.1	90.4	100.0	80.8	40.1
Panel B. Welfare: consumption-equivalent cost (% of lifetime consumption)					
Surprise shortfall	0.23	0.76	1.16	1.19	1.33
Anticipated shortfall	0.18	0.75	1.26	1.26	1.26

Note. Panel A holds behavior at the full-payment optimum and reports the fixed-behavior accounting loss. From 2034 onward, the paid-benefit fraction is 0.78, and unpaid scheduled payments are discounted at $\beta = 0.97$. Row 1 expresses the loss as a share of the present value of scheduled lifetime OASI benefits. Row 2 discounts the loss to each cohort's age 18 and divides by the present value of the cohort's age 18–80 consumption path, the same base as the welfare measure in Panel B. Row 3 discounts the loss to the common depletion year 2033 and reports an index with the largest cohort loss (the 1975 cohort) equal to 100. Panel B reports consumption-equivalent welfare costs from the permanent endpoint of the canonical duration-frontier run: surprise shortfall compares the surprise and full-payment environments, while anticipated shortfall compares the anticipated and full-payment environments. Each cohort cell contains 20,000 simulated people, with common random numbers across its matched environments. The table does not display the paired conditional Monte Carlo intervals; those intervals cover finite-simulation variation only and omit parameter, model, empirical, and multiple-comparison uncertainty. Panel A and Panel B measure different valuation concepts, so their difference is not a behavioral decomposition.

payments depends on the valuation date. Discounting every loss to 2033 makes the youngest cohort look lightly exposed even though its lifetime welfare cost is among the largest. Second, fixed-behavior accounting and the model's CEV measure different objects. The accounting rows value missing transfers along the benchmark allocation; the CEV values the complete model-implied allocation, including consumption, leisure, and choices under each environment. In the permanent experiment, the CEV is 15–38 percent below the accounting loss as a share of lifetime consumption. That gap is a descriptive comparison between valuation concepts, not the share of income "recovered" through behavior. A behavioral decomposition would require an additional welfare experiment that holds the policy environment fixed while constraining selected choices.

These results are conditional incidence, not an unconditional forecast. Every experiment fixes OASI reserve depletion at the end of 2033 and changes only expected reduced-payment duration. The model is partial equilibrium, and the financing of full-payment restoration is unspecified and excluded from welfare. The purpose is narrower: to show how payment duration and remaining adjustment capacity reallocate welfare incidence across birth cohorts.

5.6. Summary of the Duration Mechanism

Expected reduced-payment duration determines who bears the welfare cost of an OASI payment shortfall. If full scheduled payments resume quickly, the payment shortfall mainly affects households already receiving OASI or close to claiming. If reduced payments persist, they increasingly reach the retirement years of younger cohorts. The surprise-shortfall cost makes this shift clear and closely tracks mechanical benefit exposure.

Anticipatory self-insurance is a separate force. Households can save more, change work and claiming, and, with enough time, invest in human capital. These choices can make reduced payments easier to absorb, but they require earlier sacrifices of consumption or leisure. The anticipation effect therefore has no uniform sign. Total welfare costs need not increase smoothly with expected duration even though mechanical benefit exposure does.

The forty-year and permanent cases are economically distinct. At a forty-year expected duration, full-payment restoration may occur before a young cohort retires; under permanent reduced payments, that probability is zero. For the 2005 cohort, 35 percent of discounted lifetime scheduled payments are exposed in the forty-year case, compared with

100 percent under permanence. The total welfare cost rises from 0.392 to 1.261 percent. Permanence is a boundary case, not a prediction that restoration will never occur.

These results describe a deliberately limited policy experiment. The analysis fixes OASI reserve depletion at the end of 2033, the paid-benefit fraction at 0.78, and a constant annual restoration hazard. It excludes the taxes, spending changes, and price effects associated with restoration. Within these limits, a brief reduced-payment spell is primarily an older-cohort problem, a persistent spell increasingly becomes a younger-cohort problem, and anticipation can raise or lower the welfare cost.

5.7. Discussion

The model mechanism complements Luttmer and Samwick (2018). They elicit respondents' subjective distributions of future Social Security benefits and their certainty equivalents. They find a sizeable average risk premium that rises with age and interpret the age gradient as consistent with older households having fewer opportunities to mitigate benefit risk through future saving and labor supply. Their survey measure does not assign that valuation to particular behaviors. Here, saving, employment, human-capital investment, retirement, and claiming are endogenous, so the model quantifies how the available adjustment set varies with life-cycle position.

The structural analysis also exposes a second force that the age gradient alone cannot reveal: reduced-payment duration. When reserve depletion occurs late in life, households have little adjustment capacity but relatively few remaining payment years. When it precedes claiming eligibility, households retain more adjustment margins but more of their scheduled payment stream is exposed. Separating the surprise-shortfall comparison from the anticipation comparison shows how the model-implied responses vary across cohorts.[7] The contribution relative to Luttmer and Samwick (2018) is therefore an account of how life-cycle timing and payment duration distribute welfare incidence across cohorts, rather than a second estimate of an average uncertainty premium.

Two related implications follow. First, a longer adjustment horizon is not equivalent to a welfare gain: anticipatory choices can raise or lower lifetime welfare. Second, claiming early

[7] The welfare magnitudes are not directly comparable. Luttmer and Samwick (2018) report willingness to give up Social Security payments to replace a perceived payment distribution with certainty; their survey measure may also include a utility cost of worry. The CEV here compares fully specified dynamic environments and is expressed as a percentage of lifetime consumption.

cannot grandfather a household into full payments because the paid-benefit fraction applies to old and new claimants. Claiming can nevertheless shift liquidity into pre-depletion years, and onset risk lowers the expected return to delaying payments that arrive farther in the future. In this sense, the payment shortfall acts like default risk on a deferred public annuity (Geanakoplos and Zeldes 2010; Caliendo, Guo, and Hosseini 2014; Hosseini 2015) and adds a channel for earlier claiming to those in the retirement literature (Rust and Phelan 1997).

6. Conclusion

The welfare incidence of OASI reserve depletion depends on how long reduced payments last. When the shortfall is brief, the burden falls mainly on cohorts already receiving or about to receive OASI payments. As the expected duration lengthens, more of each younger cohort's retirement falls inside the shortfall, and the burden moves toward them. Permanent reduced payments are one limiting case of this duration frontier, and they place the burden on distinctly younger cohorts.

Two forces explain this pattern. The first is exposure, the share of a cohort's scheduled payments that falls in reduced-payment years, holding behavior fixed. A longer shortfall raises exposure most for younger cohorts, which is why the burden shifts toward them. The second is adjustment capacity, the time and margins a cohort has to self-insure. Households near retirement have little room to adjust. Younger households can save, work longer, invest in skills, and delay claiming, but each of these can require giving up consumption or leisure earlier in life. This is why anticipation helps some cohorts and hurts others.

This is also why unpaid dollars are an incomplete measure of welfare loss. Discounting the same unpaid payments to a different base year can reorder the accounting ranking. More fundamentally, fixed-behavior accounting and the CEV value different objects. Accounting values missing transfers along the benchmark allocation, whereas the CEV values the complete model-implied allocation, including consumption, leisure, and endogenous choices. The numerical gap between the two measures is therefore a comparison of valuation concepts, not a decomposition of income recovered through behavior.

The model is partial equilibrium with fixed prices and tax rules. The full-payment benchmark and reduced-payment paths are imposed exogenously, and the model neither specifies nor charges households for the taxes or spending changes needed to finance them. Beliefs are common within each cohort, and the reported Monte Carlo intervals condition on the calibrated parameters, empirical inputs, and model specification. Adding general-equilibrium prices, explicit financing, or belief heterogeneity within a cohort is left for future work. Still, this paper speaks to Social Security policy in two ways. First, there is no single generational burden of OASI reserve depletion. Any claim about who bears the largest welfare cost must state how long reduced payments are expected to last, because that horizon determines whether the largest modeled cost falls on cohorts near claiming or on workers still decades from retirement. Second, the same calendar event creates different model-implied welfare incidence across cohorts because exposure and adjustment capacity vary with life-cycle position. Expected payment duration and age at depletion must therefore be considered together when comparing generations.

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Appendix A. Empirical Motivation: HRS Sample

TABLE A.1. Summary statistics for the HRS estimation sample

	Mean	SD	Median
A. Demographics			
Age	56.3	2.9	56
Female	59.3%		
Years of education	13.7	2.5	14
Black	12.9%		
Hispanic	8.0%		
Married or partnered	77.7%		
Fair or poor health	16.0%		
B. Economic variables			
Currently employed	79.8%		
Has pension on current job (among employed)[d]	58.8%		
Annual earnings ($000, 2004 dollars)	35.3	113.7	24.4
Nonhousing wealth ($000, 2004 dollars)	94.1	323.0	9.6
Liquid wealth ($000, 2004 dollars)	24.2	90.2	4.5
IRA balance ($000, 2004 dollars)	72.2	342.9	0.0
C. Beliefs and expectations			
Perceived personal benefit risk (%)	60.8	28.2	60
Perceived general program risk (%)	68.1	24.1	75
Subjective probability of surviving to 75 (%)[a]	65.7	27.1	75
Perceived chance of a major economic depression (%)[b]	47.2	27.7	50
Expected probability of working after 62 (%)	47.2	37.8	50
Expected probability of working after 65 (%)[c]	30.2	33.7	20

Note. N = 3,359 except where noted: ^{a}N = 3,289, ^{b}N = 3,307, ^{c}N = 3,354; these three items have limited additional survey nonresponse within the estimation sample. ^{d}N = 2,665, the employed subsample; pension coverage on the current job is undefined for respondents who are not currently working, so the share is computed only among the employed rather than treating non-workers as lacking a pension. Earnings, nonhousing wealth, liquid wealth, and IRA balance are right-skewed; the median is a more representative summary than the mean for those rows. A median IRA balance of 0.0 reflects that most respondents report no IRA holdings. Liquid wealth and IRA balance are components of nonhousing wealth and do not sum to it exactly because nonhousing wealth includes other financial assets. Binary variables report only the sample share.

Source. Author's calculations from the 2006 HRS Internet Survey.

TABLE A.2. Belief-to-intended-work regressions across specifications

	(1)	(2)	(3)	(4)
Panel A. Work after 62 ($W_{i,62}$)				
Perceived personal benefit risk (R_i^P)	0.830***	0.858***	0.915***	0.837***
	(0.277)	(0.281)	(0.279)	(0.267)
General program risk (R_i^G)	−0.318	−0.534*	−0.501	−0.278
	(0.315)	(0.320)	(0.316)	(0.305)
Observations	3,359	3,243	3,222	3,342
R^2	0.110	0.132	0.152	0.174
Panel B. Work after 65 ($W_{i,65}$)				
Perceived personal benefit risk (R_i^P)	0.773***	0.783***	0.838***	0.737***
	(0.252)	(0.255)	(0.254)	(0.245)
General program risk (R_i^G)	−0.423	−0.602**	−0.551*	−0.347
	(0.286)	(0.289)	(0.289)	(0.280)
Observations	3,362	3,249	3,225	3,345
R^2	0.077	0.104	0.108	0.123
Age fixed effects	Yes	Yes	Yes	Yes
Demographic and 2006 wealth/earnings controls	Yes	Yes	—	Yes
Subjective survival and depression expectations		Yes		
2004 (pre-belief) controls, incl. 2004 work status			Yes	
Pension coverage on current job (3-category)				Yes

Note. Each column reports coefficients from equation (1) estimated jointly on perceived personal benefit risk (R_i^P) and general program risk (R_i^G), both in units of 10 percentage points. Standard errors, clustered by household, are in parentheses. Column (1) is the specification used in Figure 1 and reported in the main text: age fixed effects, gender, education, race and ethnicity, marital status, self-reported health, and inverse-hyperbolic-sine-transformed nonhousing wealth and earnings, all measured in 2006. Column (2) adds the person's subjective probability of surviving to 75 and perceived chance of a major depression. Column (3) replaces the 2006 demographic and financial controls with their 2004 (pre-belief) counterparts, plus 2004 employment status and hours, to check whether the relationship reflects conditions already in place before the belief was elicited. Column (4) adds pension coverage on the current job as a three-category control (no pension, has a pension, or not currently working), since pension coverage is undefined for non-workers. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Source. Author's calculations from the 2006 (and, for column (3), 2004) HRS Internet Survey.

TABLE A.3. Predictive content of stated work plans for later full-time employment

	2008 (ages 60–61 in 2006)			2010 (ages 58–61 in 2006)		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable	$W_{i,62}$	Full-time work	Full-time work	$W_{i,62}$	Full-time work	Full-time work
Perceived personal benefit risk (R_i^P, per 10pp)	1.110*	0.240	0.020	1.604***	0.291	−0.189
	(0.585)	(0.661)	(0.669)	(0.432)	(0.503)	(0.481)
General program risk (R_i^G, per 10pp)	0.773	0.446	0.124	0.002	0.948	0.762
	(0.697)	(0.795)	(0.795)	(0.507)	(0.589)	(0.558)
Stated plan: work after 62, $W_{i,62}$ (per 1pp)			0.298***			0.369***
			(0.055)			(0.038)
2006 full-time work status		Yes	Yes		Yes	Yes
Household controls[a]	Yes	Yes	Yes	Yes	Yes	Yes
Observations	601	567	559	1,254	1,129	1,119
R^2	0.175	0.506	0.549	0.139	0.259	0.329

Note. Sample restricted to respondents who were ages 60–61 (columns 1–3) or 58–61 (columns 4–6) in 2006, so that all respondents have reached age 62 by the outcome year (2008 or 2010, respectively). Columns (1) and (4) regress the stated probability of working full time after 62, $W_{i,62}$ (0–100 scale), on the two risk measures and household controls; coefficients are already in percentage points of the stated probability. Columns (2)/(5) and (3)/(6) are linear probability models for full-time work in 2008/2010. They include the same regressors and 2006 full-time work status; columns (3) and (6) additionally include the 2006 stated plan $W_{i,62}$. Coefficients on the risk measures and $W_{i,62}$ are multiplied by 100 and expressed in percentage points. Standard errors, clustered by household, are in parentheses. [a] Same controls as column (1) of Table A.2: age fixed effects, gender, education, race and ethnicity, marital status, self-reported health, and inverse-hyperbolic-sine-transformed nonhousing wealth and earnings, all measured in 2006. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Source. Author's calculations from the 2006, 2008, and 2010 HRS.

Appendix B. Measuring the OASI Financial-Risk Index

This appendix describes the construction of the empirical subjective-onset-hazard series. The input history is monthly from January 1995 through March 2026. I construct parallel OASI and OASDI series, but the paper reports OASI as its headline measure.

B.1. Information set and timing

Table B.1 lists the four blocks. All inputs are oriented before filtering so that a larger value means more adverse OASI financial conditions. The three trust-fund variables come from the intermediate-assumption OASI projections in successive SSA Trustees Reports. A report enters the filter only in its release month. In other months, the trust-fund variables are missing in the measurement equation, although tables and figures carry forward the most recently published values.

TABLE B.1. Variables used in the OASI financial-risk index

Block	Public series	Oriented measurement
Trust-fund level	75-year actuarial balance, open-group unfunded obligation, and projected intermediate-assumption reserve-depletion year	Actuarial deficit, unfunded obligation, and inverse years to depletion
Macro stress	BLS total nonfarm payroll employment, average hourly earnings, and unemployment, obtained from FRED	Unemployment and negative year-over-year growth in the product of employment and average hourly earnings
Official uncertainty	Successive SSA Trustees Reports	Absolute changes in the three trust-fund measures between consecutive Trustees Reports
Economic-policy uncertainty	Baker, Bloom, and Davis (2016) monthly US news-based Economic Policy Uncertainty index	Log EPU, released to the index construction after the source's revision window closes

Note. The three trust-fund components receive equal within-block weights, as do the two macro-stress components. The headline weights are 0.30, 0.20, 0.10, and 0.40 in the order shown. Official text is excluded. Source: Author's calculations from SSA Trustees Reports, BLS, FRED, and the Baker, Bloom, and Davis (2016) US Economic Policy Uncertainty index.

Each input enters the construction only after it would have become public. Employment, earnings, and unemployment for month t enter in month $t+1$, matching the BLS release lag. The paper uses latest-revised FRED histories shifted by one month rather than ALFRED first-release vintages. It therefore reproduces publication timing but not the data vintage available in real time; the one-sided history is an approximation to a real-time information set. The EPU index for month t can be revised in months $t+1$ and $t+2$ as newspaper archives fill in, so it enters in month $t+3$.

B.2. Transformations and measurements

Constructing the index takes two steps. This subsection turns each raw input into a standardized block measurement, one number per block per month. The next subsection filters those measurements over time and combines the four blocks into a single index.

Let m_{jt} denote component j in the month t it becomes available to the index construction, signed so that a larger value means worse OASI finances. The actuarial deficit is minus the reported actuarial balance. Depletion proximity is one over the number of years from the report year to the projected depletion year. Wage-bill stress is minus the year-over-year growth in the product of payroll employment and average hourly earnings.

Each component is standardized in two steps. I take logs of the three positive trust-fund components and the EPU index, leaving the others in levels, so that $m^*_{jt} = \log m_{jt}$ for those four series and $m^*_{jt} = m_{jt}$ otherwise. I then compute

$$z_{jt} = \frac{m^*_{jt} - \mu_j}{\sigma_j}, \qquad x_{jt} = \frac{\operatorname{asinh}(z_{jt}) - \mu^a_j}{\sigma^a_j},$$

where μ_j and σ_j are the mean and standard deviation of m^*_{jt}, and μ^a_j and σ^a_j are the mean and standard deviation of $\operatorname{asinh}(z_{jt})$.

The first step puts each component in standard-deviation units. The inverse hyperbolic sine then compresses extreme values without changing their order, so a ten-standard-deviation spike counts about three times, not ten times, a one-standard-deviation move. Because this compression changes the scale, the second step re-standardizes so that every component again has mean zero and unit variance, and equal weights within a block correspond to equal contributions. All means and standard deviations are computed once from observations until December 2011, the last HRS survey year used for calibration, and are held fixed.

The four block measurements are then simple averages of transformed components. Trust-fund level is the mean of transformed actuarial deficit, unfunded obligation, and depletion proximity. Macro stress is the mean of transformed unemployment and wage-bill stress. Official uncertainty is the mean absolute change in the three transformed trust-fund components between consecutive Trustees Reports, passed through the same fixed transformation. The EPU block is the transformed, lagged EPU index. If a component is missing, it enters as zero, its calibration mean, and the weights are not rescaled. A block is missing only when all of its components are missing.

B.3. Kalman-filtering and aggregation

The four blocks of an index are not observed at the same frequency. Trust-fund and official-uncertainty measurements change only when SSA releases a new Trustees Report, roughly once a year. Macro stress and EPU are observed every month. Filtering addresses two problems created by this timing issue. It carries a block forward between releases and, when a release arrives, updates the block gradually rather than all at once.

I filter each block measurement M_t^k separately, where k indexes the trust-fund, macro-stress, official-uncertainty, and economic-policy-uncertainty blocks. The construction therefore uses four univariate filters rather than one multivariate factor model.[8] Each block has a latent state F_t^k that evolves as

$$F_t^k = \phi_k F_{t-1}^k + \eta_t^k, \qquad \mathrm{Var}(\eta_t^k) = q_k,$$

and that is observed with noise,

$$M_t^k = H_k F_t^k + \varepsilon_t^k, \qquad \mathrm{Var}(\varepsilon_t^k) = R_k.$$

The persistence ϕ_k governs how much of the state carries from one month to the next, and q_k is the variance of the state innovation. The loading H_k converts state units into measurement units, and R_k is the variance of measurement error. I fix ϕ_k, q_k, H_k, and R_k as calibration choices rather than estimate them. The recursions are the standard scalar linear state-space updates (Hamilton 1994; Durbin and Koopman 2012).

[8] This is not merely a computational simplification. In a single multivariate filter, the common state would be updated only by the series observed in a given month, so in the eleven months between Trustees Reports the macro and EPU series would receive all effective weight. Filtering the blocks separately and aggregating with fixed weights keeps the stated weights operative in every month, and it lets each block carry its own persistence.

The preceding subsection already reduces each block to a single standardized number per month. The measurement and the state are therefore in the same units, and $H_k = 1$ for each k. Setting $R_k = 1$ is a normalization rather than a claim about how noisy the data are. With a scalar state and measurement, the filter depends on the prior variance, state innovation variance q_k, and measurement variance only through the ratios $P^k_{1|0}/R_k$ and q_k/R_k. Scaling all three by a common constant leaves every filtered state unchanged, so fixing $R_k = 1$ fixes units only. Conditional on the fixed persistence and initialization, q_k governs how much weight a new observation receives.

The filter is a recursion of a point estimate of the block state and the variance of that estimate. I write $F^k_{t|s}$ for the estimate of F^k_t formed from measurements dated no later than month s, and $P^k_{t|s}$ for the variance of that estimate. The pair $t \mid t-1$ therefore describes beliefs held before month t's measurement is seen, and $t \mid t$ describes beliefs held after. Only $s \leq t$ appears anywhere in the construction, which is the sense in which the filter is one-sided. The recursion begins in January 1995 at $(F^k_{1|0}, P^k_{1|0}) = (0, 1)$. The state starts at zero because every block measurement is standardized. The prior variance equals the measurement variance, so the initial belief carries the same weight as a single observation would.

In every month, I form a prediction. Taking the conditional mean and variance of the state equation gives

$$F^k_{t|t-1} = \phi_k F^k_{t-1|t-1}, \qquad P^k_{t|t-1} = \phi_k^2 P^k_{t-1|t-1} + q_k.$$

The state decays toward zero at rate ϕ_k. The variance evolves in two offsetting ways: the factor ϕ_k^2 shrinks it, while the state innovation variance $q_k = 0.10$ adds to it. The second effect dominates at the values used here, so an additional month without a release always leaves the block state less precise than the month before.

When M^k_t is observed, I update:

$$K^k_t = \frac{P^k_{t|t-1}}{P^k_{t|t-1} + 1}, \qquad F^k_{t|t} = F^k_{t|t-1} + K^k_t \left(M^k_t - F^k_{t|t-1} \right), \qquad P^k_{t|t} = (1 - K^k_t) P^k_{t|t-1}.$$

The Kalman gain K^k_t is the weight placed on the measurement innovation, $M^k_t - F^k_{t|t-1}$. When the prior variance $P^k_{t|t-1}$ is large relative to measurement variance $R_k = 1$, the update moves farther toward the new measurement. When the prior variance is small, the update moves less. Incorporating the observation also reduces the filtered variance.

When no measurement arrives, the update is skipped and $(F^k_{t|t}, P^k_{t|t}) = (F^k_{t|t-1}, P^k_{t|t-1})$. This step is what reconciles the annual and monthly release calendars. Between two Trustees Reports the trust-fund state is carried forward unchanged while its variance accumulates at q_k per month, so by the time the next report appears the filter has become substantially more willing to move. At the parameters used here, an annually released block moves about 0.65 of the way toward a new reading, while a monthly block moves about 0.26 of the way. A once-a-year Trustees Report therefore has a large effect at release, and a single monthly print has a small one, without either frequency being rescaled by hand. This is what makes the annual and monthly blocks comparable within one index.

The persistence parameters carry the substantive assumption. I set $\phi_k = 1$ for trust-fund level and official uncertainty, so the construction holds those states fixed between Trustees Report releases. I set $\phi_k = 0.98$ for macro stress and EPU, which implies a half-life of about 34 months. A labor-market or policy-uncertainty spike therefore decays in the index, while a published revision to program finances remains until the next report.

Let w_k denote the headline weight of block k,

$$w_{TF} = 0.30, \qquad w_{MS} = 0.20, \qquad w_{OU} = 0.10, \qquad w_{EPU} = 0.40,$$

which are fixed, nonnegative, and sum to one. I standardize each filtered block by its January 1995–December 2011 mean and population standard deviation, yielding Z^k_t. The monthly state used for the HRS mapping is the weighted sum

$$G_t = \sum_k w_k Z^k_t.$$

Conditional on the staged input history, later observations do not revise an earlier G_t.

The figures rescale G_t to a readable level; the HRS mapping uses the unrescaled state. All display moments are taken over January 1995–December 2019, a benchmark window that excludes COVID-19. Write μ_G and σ_G for the mean and population standard deviation of G_t over that window, and μ_{Z^k} for the mean of Z^k_t over the same window. The headline index and the contribution of block k are

$$I^H_t = 100 + 10\,\frac{G_t - \mu_G}{\sigma_G}, \qquad C^k_t = 10\,w_k\,\frac{Z^k_t - \mu_{Z^k}}{\sigma_G},$$

so that the contributions add up to the index exactly, $I_t^H - 100 = \sum_k C_t^k$.

The decomposition figure groups the four contributions into a fiscal and an uncertainty index,

$$I_t^F = 100 + \frac{C_t^{TF} + C_t^{MS}}{0.70}, \qquad I_t^U = 100 + \frac{C_t^{OU} + C_t^{EPU}}{0.30},$$

which gives $I_t^H = 0.70 I_t^F + 0.30 I_t^U$ exactly. The constants 0.70 and 0.30 scale the two displayed indexes and do not replace the block weights w_k; any pair summing to one reproduces I_t^H. Before smoothing, all three indexes have a January 1995–December 2019 mean of 100. Both index figures display a one-sided 12-month trailing mean, which neither the HRS mapping nor the structural model uses. The annual state G_y is the calendar-year mean of the unsmoothed monthly OASI values of G_t.

B.4. Mapping the index to HRS beliefs

The HRS Internet Surveys provide the probability scale. I use the general Social Security expectations item in 2006, 2007, and 2011. It asks respondents, thinking about Social Security in general rather than only their own benefits, for the percent chance that Congress will change the program within the next 10 years so that it becomes less generous. The wording is essentially constant across waves; in 2011 the item appears in Collection A of a randomized wording experiment. Valid answers lie between zero and 100 and are divided by 100.

The response is a proxy for belief about a future policy-induced reduction in generosity, not an objective forecast of the statutory depletion date. The Internet Surveys also contain personalized questions. In 2006 and 2007, current recipients report the chance that current benefits will be cut, while nonrecipients report the chance that changes will reduce future benefits. The 2011 personalized follow-up is conditional on a positive answer to the general question and, for nonrecipients, a positive chance of ever receiving Social Security; it is restricted to respondents younger than 70. The 2011 survey also fields an alternative general "benefits will be cut" wording. These personalized and alternative-wording items are diagnostic alternatives and do not enter the reported calibration.

Let G_y be the annual state and define

$$\lambda_y = \Lambda(\alpha_0 + \alpha_1 G_y), \qquad \Lambda(z) = \frac{\exp(z)}{1 + \exp(z)}.$$

Because the survey asks about a 10-year horizon, the fitted survey counterpart is

$$P_y^{10}(\alpha_0, \alpha_1) = 1 - [1 - \Lambda(\alpha_0 + \alpha_1 G_y)]^{10}.$$

The public replication archive contains wave-level weighted means and observation counts, but not respondent microdata. I fit the two mapping parameters by weighted nonlinear least squares, using survey-wave observation shares as weights:

$$(\widehat{\alpha}_0, \widehat{\alpha}_1) = \underset{\substack{-10 \le \alpha_0 \le 2, \\ 0 \le \alpha_1 \le 10}}{\arg\min} \sum_y \frac{N_y}{\sum_s N_s} \left[\overline{p}_y^{HRS} - P_y^{10}(\alpha_0, \alpha_1)\right]^2.$$

The objective uses only annual index states and weighted survey means for 2006, 2007, and 2011; no post-2011 index row enters it.

TABLE B.2. HRS moments used to estimate the subjective-probability mapping

Survey year	G_y	Observations	HRS 10-year risk	Model 10-year risk	Model annual λ_y
2006	−0.692	1,208	63.3%	61.7%	9.1%
2007	−0.704	1,487	60.3%	61.6%	9.1%
2011	1.552	2,180	72.1%	72.1%	12.0%

Note. The HRS entries are archived weighted means of the standard 10-year "less generous" question. The archive does not contain the respondent records needed to reconstruct the weights independently. Model 10-year risk is $1 - (1 - \lambda_y)^{10}$.

Source. Author's calculations from the HRS Internet Survey and the annual OASI financial-risk state.

The estimates are

$$\widehat{\alpha}_0 = -2.203, \qquad \widehat{\alpha}_1 = 0.135.$$

At $G_y = 0$, they imply an annual hazard of 9.9 percent and a 10-year risk of 64.9 percent. Identification is limited: two parameters are fit to three aggregate wave means. Leaving out 2006 or 2007 yields slopes of 0.151 and 0.114, respectively. Leaving out 2011 leaves two almost identical G_y values and pushes the numerical solution toward the parameter bounds ($\widehat{\alpha}_0 = 2.000$ and $\widehat{\alpha}_1 = 6.148$). I therefore use leave-one-wave-out results as a sensitivity diagnostic and do not report respondent-bootstrap standard errors from data that are not in the archive. The partial 2026 state is $G_y = 2.917$ and implies an annual hazard of 14.1 percent.

B.5. Mapping to an alternative model-belief specification

The estimated annual OASI hazard λ_y can be used as an alternative to the Trustees-informed onset beliefs in the main counterfactuals. Calendar years before the first observed year receive zero hazard. Calendar years after the last observed year use the last available hazard; in the current output this is the partial-2026 estimate. This extrapolation rule is applied separately to each birth cohort through the calendar-year-to-age mapping.

On the annual calendar, cumulative perceived incidence through year y is

$$1 - \prod_{s \leq y} (1 - \lambda_s).$$

In the alternative belief specification, the model's pre-depletion-to-reduced-payment transition in year y uses the annual hazard λ_y, not this cumulative probability. Full-payment restoration, when included, is governed by the distinct restoration hazard q_d.

Appendix C. Social Security and Tax Rules

This appendix records the fixed tax rules retained from Online Appendix C of Fan, Seshadri, and Taber (2024). Dollar amounts are in 2004 dollars. The benchmark combines the 2004 federal income and payroll tax schedules with the 1999 Retirement Earnings Test (RET). These historical rules are model primitives: the payment-shortfall experiment changes only the fraction of each scheduled OASI retirement payment that is actually paid.

C.1. Taxes and after-tax resources

The payroll tax consists of a 6.2-percent Social Security tax on earnings up to \$87,900 and an uncapped 1.45-percent Medicare tax. State income taxes are omitted. Federal taxes use head-of-household filing status, a \$3,100 personal exemption, and a \$7,150 standard deduction. Table C.1 gives the piecewise-linear map retained from FST.

TABLE C.1. After-tax income schedule in 2004 dollars

Marginal tax rate	Pre-tax income Y	After-tax income $\mathcal{T}(Y)$
0.0765	$Y \leq 10{,}250$	$0.9235Y$
0.1765	$10{,}250 < Y \leq 20{,}450$	$9{,}465.88 + 0.8235(Y - 10{,}250)$
0.2265	$20{,}450 < Y \leq 49{,}150$	$17{,}865.58 + 0.7735(Y - 20{,}450)$
0.3265	$49{,}150 < Y \leq 87{,}900$	$40{,}065.03 + 0.6735(Y - 49{,}150)$
0.2645	$87{,}900 < Y \leq 110{,}750$	$66{,}163.15 + 0.7355(Y - 87{,}900)$
0.2945	$110{,}750 < Y \leq 172{,}950$	$82{,}969.33 + 0.7055(Y - 110{,}750)$
0.3445	$172{,}950 < Y \leq 329{,}350$	$126{,}851.43 + 0.6555(Y - 172{,}950)$
0.3645	$Y > 329{,}350$	$229{,}371.63 + 0.6355(Y - 329{,}350)$

Note. The displayed marginal rate combines federal income, Social Security payroll, and Medicare taxes. Above the Social Security taxable maximum, the 6.2-percent component drops out, producing the lower fifth-row rate.

C.2. Understanding the fixed-behavior examples

These examples unpack the fixed-behavior benchmark in Figure 7. They hold behavior fixed and ask how different actual-payment paths change the value of scheduled OASI payments. The goal is not to measure life-cycle welfare. Instead, the examples show why the

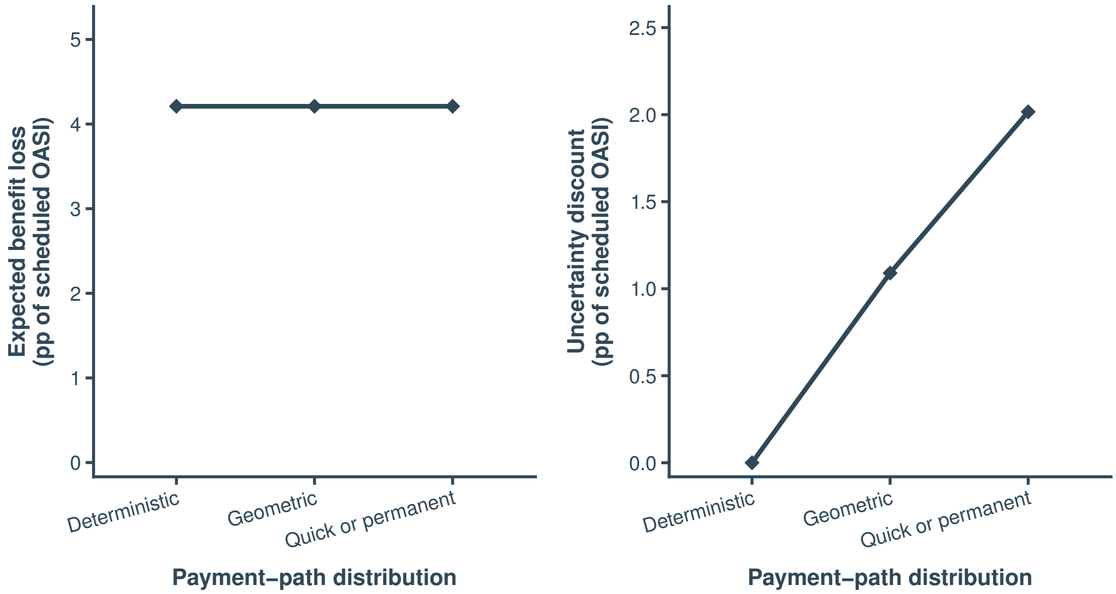

FIGURE C.1. Equal expected accounting loss does not imply equal duration risk

Note. All three policies are mean-matched for the 1982 birth cohort claiming at age 67. The calibrated geometric hazard and the quick-restoration-or-permanent mixing probability are then held fixed. Panel A verifies the equal expected loss at the reference household. Panel B reports the uncertainty discount. Values are percentage points of normalized scheduled OASI wealth.

expected accounting loss and the uncertainty discount in equation (43) measure different objects.

The same average loss can imply different uncertainty discounts. Figure C.1 considers the 1982 cohort and three full-payment-restoration processes. Each process has the same expected accounting loss: 4.2 percentage points of scheduled OASI wealth. In the first, the restoration date is known, so there is no uncertainty discount. In the second, restoration can occur in any year, so the discount is 1.1 percentage points. In the third, full payments either return quickly or never return, so the discount is 2.0 percentage points. The average loss is the same in all three cases. What changes is uncertainty about full-payment restoration.

Uncertainty about duration matters more than uncertainty about onset in this calibration. Figure C.2 separates two questions: when does reserve depletion occur, and how long do reduced payments last? The figure allows depletion to fall anywhere from 2027 through 2050 and sets expected reduced-payment duration to 20 years on average. The line for onset uncertainty alone is near zero, so almost the entire discount from both types of

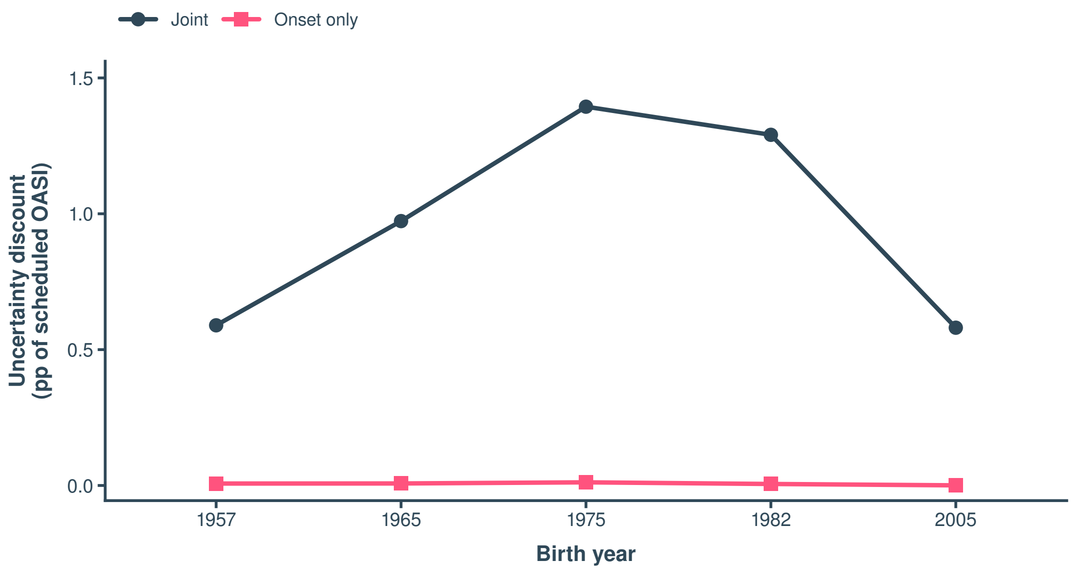

FIGURE C.2. Duration uncertainty accounts for nearly all of the uncertainty discount

Note. Reserve depletion can occur from 2027 through 2050. Full scheduled payments resume with a 5-percent annual probability, implying an expected reduced-payment duration of 20 years. The figure shows the uncertainty discount when both dates are uncertain and when only onset is uncertain. The duration-only series is not drawn because it visually coincides with the joint series. Values are percentage points of normalized scheduled OASI wealth.

uncertainty together comes from duration uncertainty. The duration-only line is omitted from the figure because it visually coincides with the joint line. A separate allocation gives duration uncertainty more than 98 percent of the total uncertainty discount for every cohort. This result depends on the assumed distributions, but it explains why the main text focuses on the length of the reduced-payment state.

Payment depth and duration determine which cohorts are exposed. Figure C.3 compares a deep but temporary payment shortfall with a smaller permanent payment shortfall. The two policies are designed to give the 1982 cohort the same expected loss, but they do not affect other cohorts in the same way. The deeper temporary shortfall is concentrated on people who receive OASI payments during 2034–2053, while the permanent shortfall reaches younger cohorts as well. Thus, matching the average loss for one cohort does not equalize losses across generations. This is the life-cycle reason that duration matters.

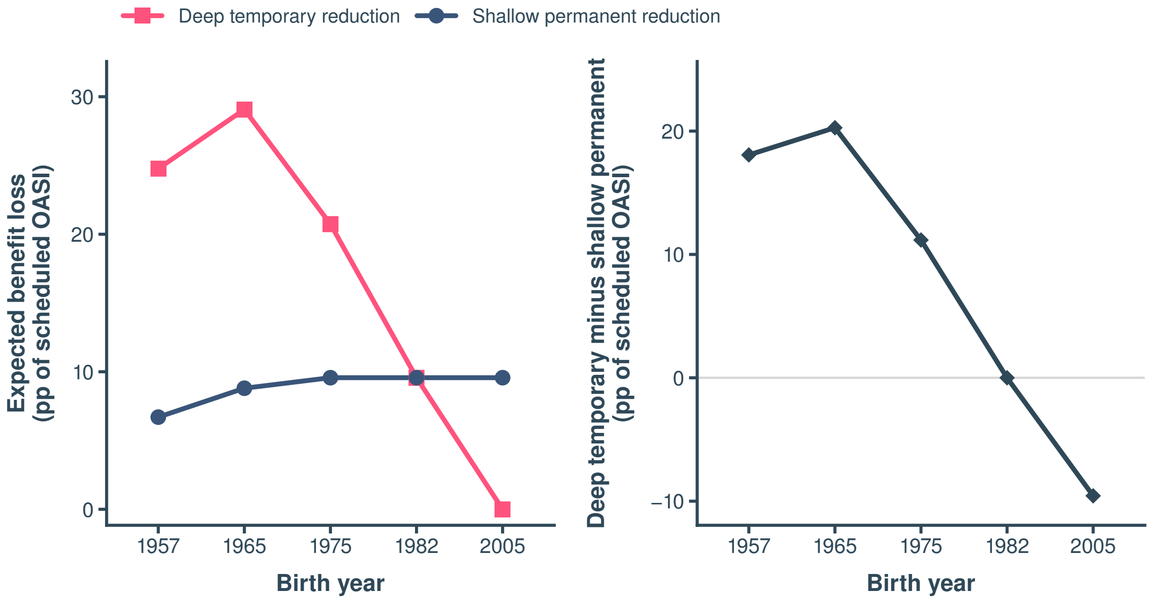

FIGURE C.3. Severity and duration redistribute the same reference loss

Note. The deep-temporary policy pays 50 percent of the scheduled amount from 2034 through 2053. The shallow-permanent policy pays 90.432 percent from 2034 onward. The policies are mean-matched only for the 1982 cohort claiming at age 67. Both policies are deterministic and therefore have zero uncertainty discount conditional on onset.

Appendix D. Model Details and Computation

This appendix completes the description of the life-cycle model in Section 3. It contains two parts. First, I derive the closed-form integration of the leisure-taste shock stated in Equation (8) and its limiting case. Second, I write out the expectation operator over next-period shocks in the Bellman equation (32) and identify the cases in which its integrals degenerate. The economic environment is that of Fan, Seshadri, and Taber (2024); labor supply is binary throughout, so part-time work is outside the model.

D.1. Analytic integration of the leisure-taste shock

Section 3 defines the conditional branch values $W_t^W(\widetilde{X}_t)$ and $W_t^N(\widetilde{X}_t)$, each optimized over consumption, claiming, and (for the working branch) investment, and each excluding the current leisure payoff $\exp(\mu_{it} + \varepsilon_{it})$. When $W_t^W > W_t^N$, the household works if and only if $\varepsilon_{it} < \varepsilon_t^*$, with the cutoff given by Equation (6), and the work probability is $\Phi(q_t)$ with $q_t = \varepsilon_t^*/\sigma_\varepsilon$ as in Equation (7).

The value of the state $\widetilde{X}_t$ before the shock is realized is the expectation of the maximum over the two branches. Splitting the expectation at the cutoff,

$$E[V_t \mid \widetilde{X}_t] = \Pr(\varepsilon < \varepsilon_t^*)\, W_t^W + \Pr(\varepsilon \geq \varepsilon_t^*)\, W_t^N + E\Big[e^{\mu_{it}+\varepsilon_{it}}\mathbf{1}\{\varepsilon_{it} \geq \varepsilon_t^*\}\Big]\,.$$

The last term is the mean of a log-normal random variable truncated from below. Completing the square in the normal density gives the standard result

$$E\Big[e^{\mu_{it}+\varepsilon_{it}}\mathbf{1}\{\varepsilon_{it} \geq \varepsilon_t^*\}\Big] = \exp\left(\mu_{it} + \frac{\sigma_\varepsilon^2}{2}\right)\Phi(\sigma_\varepsilon - q_t)\,,$$

so that the pre-shock expected value has the closed form

$$E[V_t \mid \widetilde{X}_t] = \Phi(q_t)\, W_t^W + [1 - \Phi(q_t)]\, W_t^N + \exp\left(\mu_{it} + \frac{\sigma_\varepsilon^2}{2}\right)\Phi(\sigma_\varepsilon - q_t)\,. \tag{D.1}$$

This is the operator $\mathcal{E}_t$ in Equation (8). When $W_t^W \leq W_t^N$, the limiting expression as $q_t \to -\infty$ applies: the household never works and

$$E[V_t \mid \widetilde{X}_t] = W_t^N + \exp\left(\mu_{it} + \frac{\sigma_\varepsilon^2}{2}\right), \tag{D.2}$$

which is the nonwork value plus the unconditional mean of the log-normal taste shifter.

Two computational implications follow. First, because the shock is integrated exactly, no quadrature over ε_{it} is required and ε_{it} never appears in the state vector; the two conditional values W_t^W and W_t^N are sufficient statistics for the work margin. Second, Equation (D.1) combines the two conditional branch values analytically. The payment-regime extension changes the continuation values inside W_t^W and W_t^N but leaves this integration step unchanged.

D.2. Expectation operator in the Bellman equation

Conditional on a work branch $p \in \{0, 1\}$ and the associated choices of consumption, investment, and claiming, the continuation value in Equation (32) is the expectation over three sources of next-period uncertainty: the family transition, next-period spousal earnings, and the human-capital innovation. Writing $F_{\zeta,t+1}$ for the spousal earnings distribution and F_ξ for the distribution of the Ben–Porath innovation ξ^H in Equation (12), the expectation operator applied to the state $\boldsymbol{X}_t$ of Equation (31) is

$$\mathbb{E}_t\Big[\overline{V}_{t+1}\Big] = \sum_{M'} P_t^M(M_t, M') \iint \overline{V}_{t+1}(A', H', R', s', M')\, dF_{\zeta,t+1}\, dF_{\xi}, \tag{D.3}$$

where $\overline{V}_{t+1}$ is the regime-mixed continuation value of Equation (33). Two of these integrals collapse in identifiable cases. The spousal-income integral is degenerate at zero income unless the next-period family state is married-working ($M' = 2$); when $M' = 2$, spousal log earnings are log-normal with age-specific mean and variance and the integral is evaluated with two Gauss–Hermite nodes. The human-capital integral is degenerate when the household does not work: with $p_t = 0$, Equation (12) reduces to deterministic depreciation $H_{t+1} = (1 - \delta)H_t$, and the innovation is irrelevant. The idiosyncratic expectation in equation (D.3) contains no survival, medical-expenditure, job-offer, previous-employment, or separate wage shock. The current payment regime also determines current OASI payments and resources. The regime transition and belief group enter through $\overline{V}_{t+1}$ and add no dimension to the shocks integrated in (D.3).

D.3. Numerical implementation

I solve the model by backward induction from age 80, when the continuation value is the bequest function in equation (30). Assets, human capital, and the Social Security record are discretized on age-varying grids. Parameter search uses $(N_A, N_H, N_R) = (18, 16, 11)$; the reported counterfactuals use $(31, 16, 21)$. The record grid represents AIME before claiming and the claiming-age-adjusted entitlement after claiming, consistent with equation (14).

At each state and age, I solve the work and nonwork branches separately and compare the admissible claiming choices. Conditional on the discrete choices, I optimize consumption and, in the work branch, human-capital investment with a Nelder–Mead simplex routine. I evaluate the consumption-floor boundary separately to retain corner solutions.

I integrate the human-capital innovation with five-node Gauss–Hermite quadrature, spousal log earnings with two-node Gauss–Hermite quadrature, and family status with a finite Markov sum. I use tetrahedral interpolation for continuation states between grid points. The analytically integrated leisure shock combines the work and nonwork branch values using equation (D.1).

Within every branch, I update AIME before claiming, convert the record to the claiming-adjusted entitlement at initiation, apply the earnings test and its future-entitlement credit, and then apply the payment factor $\chi(z_t)$. I calculate taxable benefits from the payment actually received and apply the after-tax-income map last. This order ensures that the reduced-payment state changes actual OASI payments but not AIME, the claiming adjustment, or the earnings-test credit. The adjusted entitlement is interpolated on the common record grid after claiming. All environments use the same grid and numerical settings.

Appendix E. Additional life-cycle results

This appendix provides complete life-cycle profiles for the permanent reduced-payment endpoint. The main text connects expected reduced-payment duration to mechanical benefit exposure, welfare incidence, and anticipatory adjustment. The material below reports the underlying cohort profiles.

E.1. Surprise-shortfall adjustment is age dependent

The surprise-shortfall effect in equation (36) measures the model-implied consequence of an unexpected decline in actual OASI payments. In Figure E.1, each cohort follows the full-payment benchmark through 2033; all later differences are ex-post behavioral responses. Reduced payments raise the marginal value of resources, but life-cycle position determines which adjustment margins remain available.

For the 1957 cohort, reduced payments begin at age 77, after claiming is complete. Actual OASI payments fall by 1.41 model units at that age, consumption falls by 0.087 units, and assets end 3.67 units below the benchmark at age 80. Claiming cannot move, human-capital adjustment is negligible, and employment rises by at most 0.49 percentage points. The old-age response is therefore almost pure dissaving: households absorb part of the lost transfer by consuming less and running down accumulated assets. They have little ability to self-insure after the surprise, but they also lose payments for relatively few remaining years; the surprise-shortfall welfare cost is 0.23 percent of lifetime consumption.

The 1965 cohort encounters the shortfall near the retirement transition. Reduced payments begin at age 69, when claiming is effectively complete but a substantial retirement horizon remains. Employment rises by 3.89 percentage points at onset. Consumption reaches 0.35 model units below the benchmark at age 71, and assets are 9.65 units lower by age 80. Human capital eventually rises by 0.21 units, but the short horizon limits the return to new investment. This combination—many remaining payment years but few remaining adjustment years—produces the largest point-in-time consumption and asset losses among the three cohorts discussed, even though its lifetime welfare loss, 0.76 percent, is not the largest.

The 1975 cohort enters the reduced-payment state at age 59, before becoming eligible to claim, and therefore retains a broader set of adjustment margins. The share that has

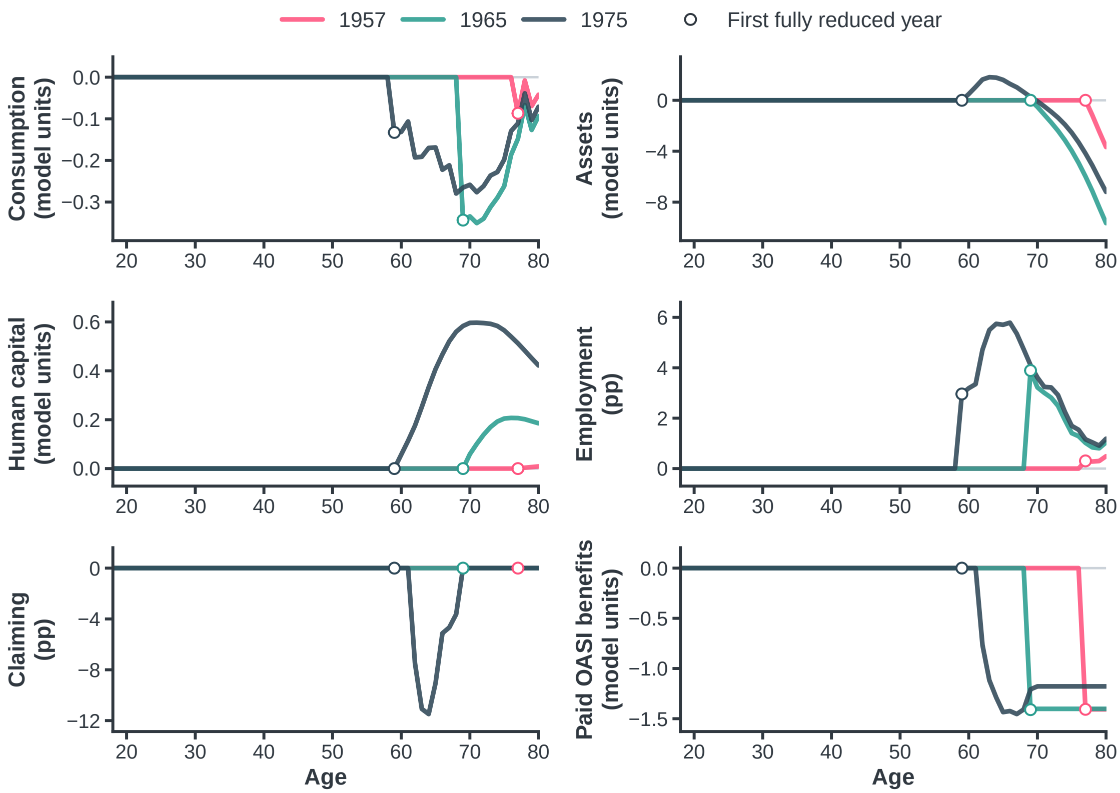

FIGURE E.1. Life-cycle responses to a surprise OASI payment shortfall

Note. Lines show the 1957, 1965, and 1975 birth cohorts, which are ages 76, 68, and 58 in the 2033 depletion year and ages 77, 69, and 59 in the first reduced-payment year. These cohorts encounter reduced payments after, during, and before the claiming window, respectively. The figure plots the surprise-shortfall environment minus the full-payment FST-based benchmark at each age. Employment and claiming differences are percentage points; all other differences are in model units. Open circles mark the first reduced-payment age. Structural parameters, initial conditions, and simulation draws are common across scenarios.

claimed by age 64 is 11.49 percentage points lower than in the benchmark. Employment rises by as much as 5.79 percentage points at age 66, and the longer expected work horizon raises the return to training: human capital peaks 0.60 model units above the benchmark at age 71. Assets initially rise by 1.82 units at age 63 and then fall to 7.19 units below the benchmark by age 80 as the accumulated buffer replaces part of the missing payment income. Because the cohort can adjust work, productivity, saving, and claiming jointly, its maximum consumption decline, 0.28 model units, is smaller than that of the 1965 cohort.

Two objects organize these responses. Mechanical benefit exposure records how much of a cohort's scheduled payment stream falls in reduced-payment years. Adjustment capacity records which self-insurance margins remain available when the surprise occurs.

Panel A of Table E.1 reports the surprise-shortfall welfare cost. It rises from 0.23 percent for the 1957 cohort to 0.76 percent for 1965 and 1.16 percent for 1975 as younger cohorts face more remaining years of reduced payments. The largest short-run consumption decline is nevertheless non-monotone in age because households gain more adjustment margins when the event arrives before retirement. The common payment loss therefore propagates differently through retirement, asset accumulation, and human-capital investment at different ages.

E.2. Anticipation changes saving and claiming

Equation (37) measures the anticipation effect by comparing the anticipated-shortfall environment Y^A with the surprise-shortfall environment Y^U under identical objective payment histories. The asset profiles reveal precautionary saving as one adjustment margin.

At payment-shortfall onset, assets in the anticipated environment are 3.71, 9.45, and 6.55 model units higher for the 1957, 1965, and 1975 cohorts. The additional buffer raises consumption at onset by 0.033, 0.092, and 0.089 units and reduces the employment response by 0.15, 2.28, and 1.54 percentage points relative to surprise. Anticipation shifts part of the response to earlier ages.

Building these buffers requires earlier adjustment. Relative to surprise, the 1965 and 1975 cohorts consume 0.075 and 0.076 model units less at earlier ages. The 1965 cohort builds the largest asset buffer against a nearby retirement-income loss. The 1975 cohort has more time to self-insure through human-capital investment and has 0.14 model units more human capital at onset. For the 1957 and 1965 cohorts, the remaining horizon is too short for marginal human-capital investment to provide much insurance.

Claiming also shifts resources across time. For the 1965 cohort, the share that has claimed by age 63 is 13.19 percentage points higher under anticipation, although cumulative claiming is equal across environments by age 69. Early claiming cannot preserve the full-payment rate because ϕ applies to old and new claimants alike. It instead moves liquidity into pre-depletion years and reduces exposure to the lower expected return on payments delayed farther into the future.

Figure E.3 shows this reallocation across claiming ages. Positive values indicate that anticipation moves applications into an age, while negative values indicate that it moves applications away from that age. For the 1965 cohort, the increase at the earliest claiming

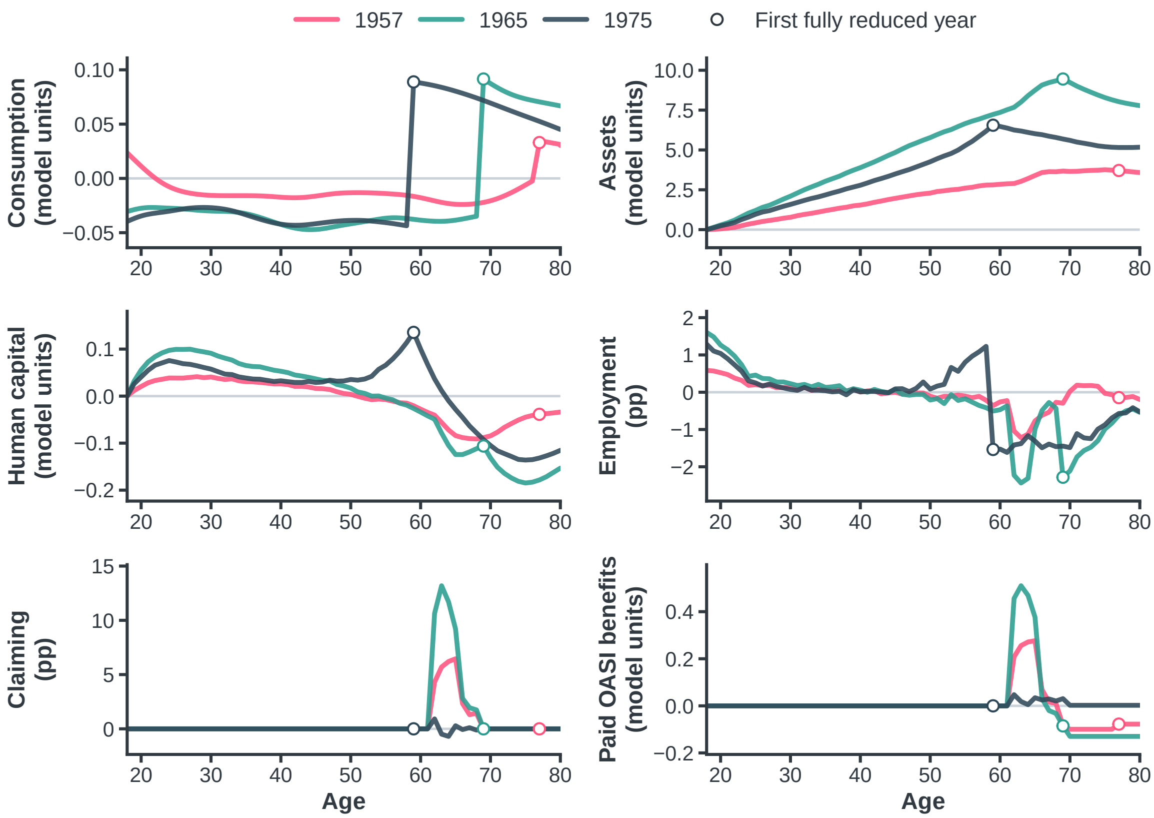

FIGURE E.2. Life-cycle adjustment under anticipation

Note. Lines show the 1957, 1965, and 1975 birth cohorts, which are ages 76, 68, and 58 in the 2033 depletion year and ages 77, 69, and 59 in the first reduced-payment year. These cohorts encounter reduced payments after, during, and before the claiming window. The figure plots the anticipated-shortfall environment minus the otherwise identical surprise-shortfall environment at each age, thereby isolating the anticipation effect. Employment and claiming differences are percentage points; all other differences are in model units.

ages is followed by fewer applications at later ages. The figure therefore shows a change in when households claim, not a lasting increase in the number of claimants.

More broadly, anticipation changes the timing of adjustment rather than eliminating the underlying loss. Households may save more, reallocate consumption across ages, work more before reduced payments begin and less when it occurs, claim benefits earlier, and—when they have a long enough horizon—invest in skills that raise future earnings. These actions can reduce the immediate drop in consumption and employment when payments fall. However, they also require earlier sacrifices in consumption or leisure, so smoother adjustment does not necessarily translate into higher welfare over the full life-cycle.

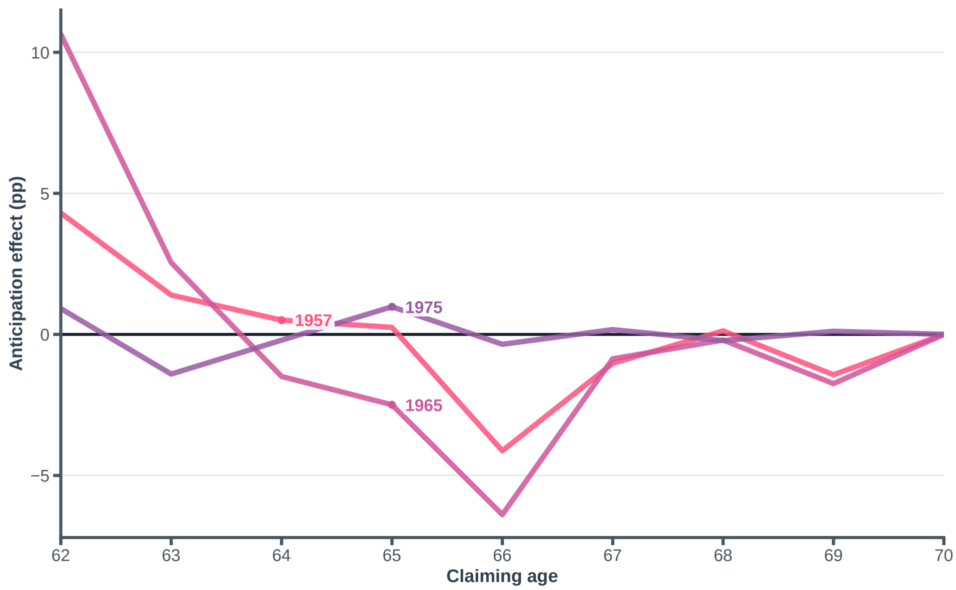

FIGURE E.3. Anticipation changes the timing of Social Security claiming

Note. Lines report the difference in application rates between the anticipated-shortfall and surprise-shortfall environments at each claiming age, in percentage points. Both scenarios face the same realized 2033 reserve-depletion event and permanent reduced payments; they differ in whether households anticipate before the event. The 1957, 1965, and 1975 birth cohorts experience the reduced payments after, during, and before the Social Security claiming window, respectively. Positive values indicate that anticipation moves applications into an age, while negative values indicate that it moves applications away from that age.

E.3. Welfare at the permanent reduced-payment boundary

Sections E.1 and E.2 report the surprise-shortfall and anticipation comparisons. The total CEV is computed directly from the anticipated-shortfall versus full-payment comparison. It is not obtained by adding the two component CEVs, because each CEV applies a nonlinear transformation to a different target–reference utility gap.

Figure E.4 plots the total effect. Total welfare costs in the canonical simulation are 0.18, 0.75, 1.26, 1.26, and 1.26 percent for the 1957, 1965, 1975, 1982, and 2005 cohorts. The costs rise across the first three cohorts and then flatten. For the three youngest cohorts, reserve depletion precedes claiming eligibility, so nearly the entire future scheduled OASI payment stream is exposed to permanently reduced payments.

The three matched comparisons clarify why total costs flatten for the three youngest cohorts. The surprise-shortfall cost rises across them, while the anticipation comparison becomes more favorable for the 2005 cohort, which has a full working life over which to spread adjustment.

Figure E.5 summarizes the employment and welfare responses across all five cohorts. The employment response changes sharply with life-cycle position. The two oldest cohorts make relatively small changes in total employment, whereas the 1975, 1982, and 2005 cohorts work about 0.62 additional years over the life-cycle. These younger cohorts have more time to respond because reserve depletion occurs well before they become eligible to claim benefits. Their welfare costs are also similar, at about 1.26 percent of lifetime consumption because nearly their entire future scheduled OASI payment stream is exposed to the payment shortfall.

Figure E.4 and Table E.1 show that life-cycle position, not distance in birth years, organizes welfare incidence. The 1965 cohort is 68 at depletion, while the 1975 cohort is 58 and not yet eligible to claim. Their total welfare costs are 0.747 and 1.257 percent. The 1975 and 2005 cohorts are thirty years apart but have similar costs of 1.257 and 1.261 percent. Once depletion precedes claiming eligibility, nearly the entire scheduled payment stream is exposed. A longer adjustment horizon still changes saving, employment, human-capital investment, and claiming, but these changes do not monotonically reduce welfare costs. Birth year is therefore not a sufficient statistic for incidence.

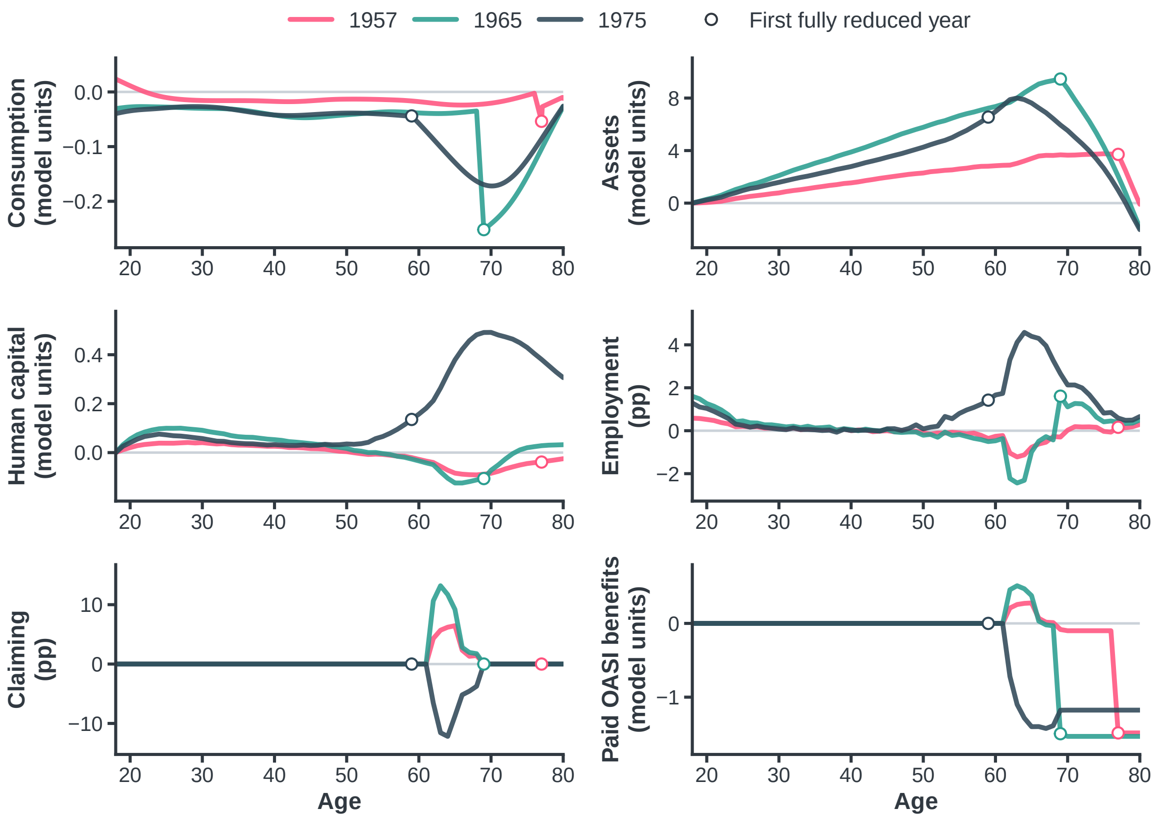

FIGURE E.4. Total life-cycle effect of the anticipated payment shortfall

Note. Lines show the 1957, 1965, and 1975 birth cohorts, which are ages 76, 68, and 58 in the 2033 depletion year and ages 77, 69, and 59 in the first reduced-payment year. These cohorts encounter reduced payments after, during, and before the claiming window, respectively. The figure plots the anticipated-shortfall environment minus the full-payment FST-based benchmark at each age. Employment and claiming differences are percentage points; all other differences are in model units. Structural parameters, initial conditions, and simulation draws are common across scenarios. Flow-outcome differences telescope across the three environments; each welfare CEV is computed separately from its stated target–reference comparison.

E.4. Details of the accounting comparison

Table 3 compares two valuation concepts using the same canonical simulated households. Panel A is a fixed-behavior accounting calculation. It values unpaid scheduled OASI payments along the full-payment benchmark path: payments from 2034 onward are multiplied by $1 - \phi = 0.22$ and discounted at $\beta = 0.97$. Panel B reports CEVs from the matched structural environments. The purpose is to compare what the concepts measure, not to treat their difference as a behavioral decomposition.

Panel A illustrates two accounting conventions. The lifetime convention discounts each

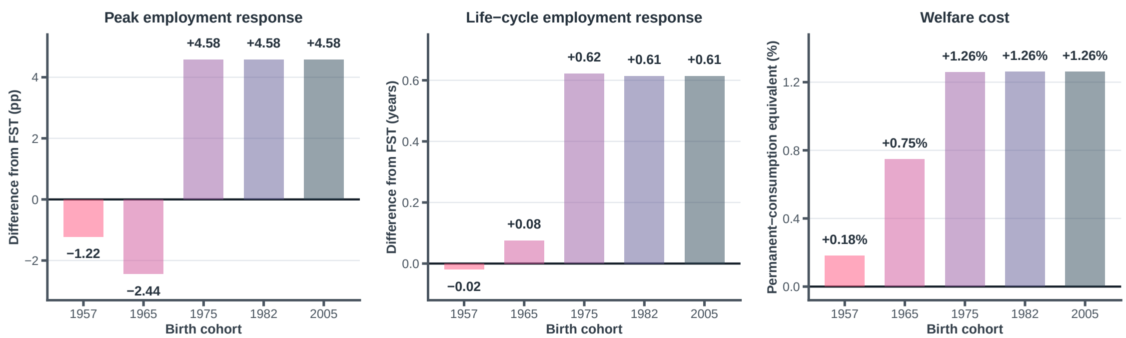

FIGURE E.5. Employment responses and welfare costs by birth cohort

Note. All outcomes compare the anticipated-shortfall environment with the full-payment FST benchmark. The left panel reports the signed age-specific employment-rate difference with the largest absolute magnitude, in percentage points. The middle panel reports the sum of age-specific employment-rate differences over the life-cycle, in years. The right panel reports the consumption-equivalent variation as a percentage of the reference scenario's age-18–80 consumption path, with welfare losses shown as positive values. All three panels report the same five birth cohorts. Reduced payments begin after the claiming window for the 1957 cohort, during it for the 1965 cohort, and before it for the 1975, 1982, and 2005 cohorts.

cohort's loss to age 18 and expresses it as a share of scheduled OASI wealth or lifetime consumption. The common-date convention discounts each cohort's unpaid payments to 2033 and normalizes the largest value to 100. These conventions answer different questions, so they need not rank cohorts in the same order.

Under the lifetime convention, the loss as a share of lifetime consumption rises from 0.29 percent for the 1957 cohort to 1.49 percent for the 1975 cohort and then plateaus. Once reserve depletion precedes the claiming window, the entire scheduled payment stream is exposed in this stationary model. Under the common-date convention, the loss instead peaks for the 1975 cohort and declines for younger cohorts: the index is 40 for the 2005 cohort and 100 for the 1975 cohort because the younger cohort's unpaid payments occur much later. The ranking therefore depends on the chosen valuation date.

The CEV supplies a third ranking because it measures a different object. The 2005 cohort has a common-date payment-loss index of 40, below the 1965 cohort's index of 90, but their total CEVs are 1.261 and 0.747 percent, respectively. The CEV values consumption, leisure, and choices in each full model environment rather than pricing only the missing transfers at an external date.

The levels should be read with the same distinction in mind. The lifetime-consumption accounting losses are 0.29, 0.99, and 1.49 percent for the 1957, 1965, and three fully exposed cohorts; the corresponding total CEVs are 0.18, 0.75, and approximately 1.26 percent. Thus, the CEVs are 15–38 percent below the accounting measures. This range is descriptive. It cannot be labeled the share "recovered" by saving, work, human-capital investment, or claiming, because Panel A holds behavior fixed while Panel B values a different object in an optimized environment. Identifying a behavioral recovery share would require an additional experiment that constrains selected choices within the same policy environment.

Separately, the three matched model comparisons clarify the welfare plateau among the youngest cohorts. Their surprise-shortfall CEVs rise from 1.157 to 1.194 to 1.334 percent, while the anticipation comparison changes from a cost of 0.104 percentage points for the 1975 cohort to a reduction of 0.075 points for the 2005 cohort. The total CEVs, computed directly rather than by adding component CEVs, are approximately 1.26 percent for all three cohorts.

Two qualifications apply. First, the full-payment benchmark assigns the same age profiles of scheduled payments and consumption to the 1975, 1982, and 2005 cohorts because the model contains no calendar-time growth; this stationarity produces the exact accounting plateau. Second, $\beta = 0.97$ is an illustrative discounting convention shared with the annuity calculation, not a claim about the trust fund's actuarial interest rate. Other discount rates would change the common-date levels and can change their relative magnitudes.

TABLE E.1. Life-cycle responses to permanent reduced OASI payments by birth cohort

	1957	1965	1975	1982	2005
Age in event year	76	68	58	51	28
Panel A. Surprise-shortfall effect					
Consumption at first reduced-payment age	-0.087	-0.343	-0.133	-0.082	-0.049
Employment at first reduced-payment age	0.305	3.890	2.960	1.180	0.265
Assets at first reduced-payment age	0.000	0.000	0.000	0.000	0.000
Human capital at first reduced-payment age	0.000	0.000	0.000	0.000	0.000
Paid OASI at first reduced-payment age	-1.407	-1.410	0.000	0.000	0.000
Consumption-equivalent welfare cost	0.229	0.756	1.157	1.194	1.334
	(0.004)	(0.014)	(0.020)	(0.020)	(0.025)
Panel B. Anticipation effect					
Consumption at first reduced-payment age	0.033	0.092	0.089	0.046	0.016
Employment at first reduced-payment age	-0.145	-2.280	-1.535	-0.940	-0.155
Assets at first reduced-payment age	3.710	9.450	6.551	4.743	1.498
Human capital at first reduced-payment age	-0.039	-0.106	0.135	0.034	0.061
Paid OASI at first reduced-payment age	-0.078	-0.085	0.000	0.000	0.000
Consumption-equivalent welfare cost	-0.049	-0.009	0.104	0.070	-0.075
	(0.030)	(0.040)	(0.035)	(0.037)	(0.032)
Panel C. Total anticipated-shortfall effect					
Consumption at first reduced-payment age	-0.054	-0.252	-0.044	-0.036	-0.032
Employment at first reduced-payment age	0.160	1.610	1.425	0.240	0.110
Assets at first reduced-payment age	3.710	9.450	6.551	4.743	1.498
Human capital at first reduced-payment age	-0.039	-0.106	0.135	0.034	0.061
Paid OASI at first reduced-payment age	-1.484	-1.495	0.000	0.000	0.000
Consumption-equivalent welfare cost	0.180	0.747	1.257	1.261	1.261
	(0.030)	(0.040)	(0.038)	(0.040)	(0.040)

Note. The common calendar event year is 2033. Panel A compares a surprise shortfall with the full-payment benchmark. Panel B compares anticipated and surprise shortfalls under the same objective payment history. Panel C compares the anticipated shortfall with the full-payment benchmark. Welfare costs are consumption-equivalent variations, reported as percentages of reference-arm consumption over ages 18–80; positive values denote losses. They are neither percentages of Social Security benefits nor realized consumption changes. Parentheses report paired conditional Monte Carlo standard errors based on common random numbers. They exclude parameter, model, empirical-sampling, and multiple-comparison uncertainty. The objective event year is fixed exogenously and is common to all environments.